Espaces de Sobolev et de Besov : réarrangements, interpolation, extrapolation

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1 Objectifs du mémoire

Dans l'écriture de ce mémoire, je me suis donné pour objectifs, d'une part, de fournir une synthèse de mon stage, entre interpolation et étude des espaces de Sobolev ; d'autre part, d'utiliser cette opportunité pour présenter certains points de vue et certaines preuves sous-representés, ou absents à ma connaissance, dans la littérature.

Les livres suivants forment le fil d'Ariane qui traverse mes lectures de stagiaire ; ils ont motivé, en grande partie, les questions qu'on trouvera dans ce mémoire. Ce sont les deux livres de

Leoni sur les espaces de Sobolev [6] et [7], ainsi que la monographie de Bennett et Sharpley [1] et celle de Bergh et Lofstrom sur l'interpolation [2].

Au gré de ces lectures, Petru Mironescu et moi formulions parfois des réserves quant aux méthodes utilisées, à l'ordre d'exposition, ou encore à la manière de motiver concrètement certaines élaborations théoriques. Il n'était pas rare qu'alors nous trouvions des approches qui nous semblassent « éclairantes », adossées à la littérature déjà fournie sur ces sujets.

Avec ce mémoire, j'aimerais, en un mot, combler certaines lacunes « folkloriques » (certains résultats sont connus de tous ; et pourtant, on a du mal à trouver les preuves ou les observations correspondantes), et présenter des méthodes que je trouve plus instructives que celles que l'on trouve dans les références sus-citées. Les sections sont indépendantes (sauf pour certaines définitions à aller chercher « plus haut »), et sont construites autour des nouvelles preuves qui constituent le cœur du mémoire. Ce sont les seules preuves du mémoire ; je fournis des références pour les énoncés orphelins.

2 Overview

Analysts use the word "interpolation" to describe a wide range of results that involve estimates on a space lying between two other spaces on a scale that can be defined, for example, by integrability or differentiability. We would like to show how this word, beyond its informal use, refers to a coherent and meaningful mathematical approach.

Let us compare the celebrated examples of the Marcinkiewicz theorem and the Gagliardo-Nirenberg inequalities in fractional Sobolev spaces. These are

Theorem 2.1. (Gagliardo-Nirenberg)
Let $1 \leq p \leq \infty$, $0 \leq s_0 < s_1 \leq 1$, and $0 \leq \sigma \leq 1$. Set $s = \dfrac{1-\sigma}{s_1} + \dfrac{\sigma}{s_1}$.
We denote $|u|_{W^{0,p}} := \|u\|_{L^p}$ exceptionally, to write an unique inequality that covers all cases. We have

$$|u|_{W^{s,p}(\mathbb{R}^N)} \leq c|u|^{1-\sigma}_{W^{s_0,p}(\mathbb{R}^N)}|u|^{\sigma}_{W^{s_1,p}(\mathbb{R}^N)} \text{ for all } u \in W^{s_1,p}(\mathbb{R}^N).$$

Theorem 2.2. (Marcinkiewicz) Let $1 \leq p_0 < p_1 < \infty$.
Let $T : L^{p_0}(\mathbb{R}^N) + L^{p_1}(\mathbb{R}^N) \to \{\text{Lebesgue measurable functions}\}$ be a linear operator such that:

- $|\{x \in \mathbb{R}^N,\ |Tf(x)| > \alpha\}| \leq K_0 \dfrac{\|f\|^{p_0}_{L^{p_0}}}{\alpha^{p_0}}$ for all $f \in L^{p_0}$.
- $|\{x \in \mathbb{R}^N,\ |Tf(x)| > \alpha\}| \leq K_1 \dfrac{\|f\|^{p_1}_{L^{p_1}}}{\alpha^{p_1}}$ for all $f \in L^{p_1}$.

Then, for all $p_0 < p < p_1$, there exists $K_p > 0$ such that

$$\|T(f)\|_{L^p} \leq K_p \|f\|_{L^p}, \text{ for all } f \in L^p(\mathbb{R}^N).$$

The proof of the later (as one finds it in [10]) was one the motivations for functional analysts to delve deeper into the description and construction of interpolation spaces. Eventually, the real and complex methods of interpolation (following, respectively, the proofs of Theorem 2.2 and of Riesz-Thorin theorem) turned out to be simple yet powerful descriptive tools, as some of the most important function spaces can be realised with those methods. The real method, for example, allows to construct a space $(X_0, X_1)_{\sigma,q}$ (see subsection 4.2) that gives consistency

to our vague intuition that most of interpolation phenomena could be related, whatever may be the (continuous) scale we manipulate.

Strikingly, as fractional Sobolev spaces and Lorentz spaces can be realized that way, we have the following results.

Theorem 2.3. (See Theorem 5.6) Let $1 \leq p \leq \infty$, $0 < s_0 < s_1 < 1$, $0 < \sigma < 1$, and set $s = \frac{1-\sigma}{s_1} + \frac{\sigma}{s_1}$. Then

$$\dot{W}^{s,p}(\mathbb{R}^N) = (L^p(\mathbb{R}^N), \dot{W}^{1,p}(\mathbb{R}^N))_{s,p}\,,$$
$$\dot{W}^{s,p}(\mathbb{R}^N) = (\dot{W}^{s_0,p}(\mathbb{R}^N), \dot{W}^{s_1,p}(\mathbb{R}^N))_{\sigma,p}.$$

Theorem 2.4. Let $1 \leq p_0 < p_1 < \infty$, $1 \leq q_0, q_1, q \leq \infty$ and $0 < \sigma < 1$. Set $p = \frac{1-\sigma}{p_0} + \frac{\sigma}{p_1}$. Then

$$(L^{p_0,q_0}(\mathbb{R}^N), L^{p_1,q_1}(\mathbb{R}^N))_{\sigma,q} = L^{p,q}(\mathbb{R}^N).$$

As a consequence of the two above theorems, one finds, for example, that theorems 2.1 and 2.2 indeed come from the same place. They are consequences of simple exercises in the abstract setting of interpolation theory.

Theorem 2.5. Let (X_0, X_1), (Y_0, Y_1) be compatible couples, and let $0 < \sigma < 1$ and $1 \leq q \leq \infty$. If $T : X_0 + X_1 \to Y_0 + Y_1$ is such that $T|_{X_i} \in B(X_i, Y_i)$ then:

$$T|_{(X_0,X_1)_{\sigma,q}} \in B((X_0, X_1)_{\sigma,q}, (Y_0, Y_1)_{\sigma,q}) \text{ with}$$
$$\|T\|_{B((X_0,X_1)_{\sigma,q},(Y_0,Y_1)_{\sigma,q})} \leq \|T\|^{1-\sigma}_{B(X_0,Y_0)}\|T\|^{\sigma}_{B(X_1,Y_1)}$$

gives Theorem 2.2, with $X_0 = L^{p_0}(\mathbb{R}^N)$, $X_1 = L^{p_1}(\mathbb{R}^N)$ and $Y_0 = L^{p_0,\infty}(\mathbb{R}^N)$, $Y_1 = L^{p_1,\infty}(\mathbb{R}^N)$ because of Theorem 2.4: $L^p(\mathbb{R}^N) = (L^{p_0,1}, L^{p_1,1})_{\theta,p} = (L^{p_0,\infty}, L^{p_1,\infty})_{\theta,p}$.

On the other hand, the following holds.

Theorem 2.6. Let (X_0, X_1) be a compatible couple and let $0 < \sigma < 1$ and $1 \leq q \leq \infty$. We have an abstract version of Gagliardo-Nirenberg inequalities with

$$\|x\|_{\sigma,q} \leq \|x\|^{\sigma}_{X_0}\|x\|^{\sigma}_{X_1}, \text{ for all } x \in (X_0, X_1)_{\sigma,q}.$$

This theorem, combined with Theorem 2.3, implies Theorem 2.1.

The examples of Theorems 2.5 and 2.6, and their applications to Theorems 2.2 and 2.1, show that an unified approach of some of the most important interpolation results is possible, using interpolation theory.

More than a way to reinterpret well-known results, this theory is a powerful tool to establish estimates in interpolation spaces, and it is particularly useful in spaces that may be difficult to manipulate in their full complexity. The general method to establish such an estimate consists in: 1) showing that the function space we are studying is an interpolation space (as in Theorems 2.4 and 2.3, for example) ; 2) proving relevant estimates in endpoint spaces ; 3) using abstract results, like Theorem 2.5 to conclude. This method allows, for example, to prove Sobolev embeddings of Besov spaces in Lorentz spaces, and other fundamental results, in a systematic way using their counterpart in endpoint spaces $W^{m,p}$ (see [6, pp.559-565]).

But interpolation theory also operates in reverse: it gives deeper insights on extrapolation, from intermediate spaces to endpoint spaces. A typical result, by Bourgain, Brezis and Mironescu, asserts the following.

Theorem 2.7. Let $f \in W^{1,p}(\mathbb{R}^N)$, $1 \leq p < \infty$. We have

$$(BBM) \quad \lim_{s \to 1^-} (1-s)^{1/p} |f|_{W^{s,p}} = K_{N,p} \|\nabla f\|_{L^p}.$$

This result is understood in the realm of so-called extrapolation theory, and it has been an important motivation for some recent developments, as it gives the idea of extrapolating fine estimates, easier to establish in fractional Sobolev spaces, to Sobolev spaces.

An important aspect of extrapolation theory is its ability to know which estimates can be extrapolated, or where one should search for those. For example, Theorem 2.7 shows that the following can be extrapolated : for every $0 < s < 1, 1 \leq p < \infty$ such that $N > sp$

$$(S) \quad \|u\|^p_{L^{\frac{Np}{N-sp}}}(\mathbb{R}^N) \leq C_{N,p} \frac{s(1-s)}{(n-sp)^{p-1}} \|u\|^p_{W^{s,p}(\mathbb{R}^N)}, \text{ for all } u \in W^{s,p}.$$

As often when keeping track of best constants, an approach exploiting symmetries turns out to be useful to prove (S). Maz'ya gave an elementary proof of (S) in [8, p.544], starting from the basic observation that we only need to prove it for radial, and radially non-increasing functions. Indeed, one can always replace a function in a fractional Sobolev space by its spherically symmetric rearrangement when studying embeddings of Sobolev spaces in Lorentz spaces, as the next theorem shows.

Theorem 2.8. Let $1 \leq p < \infty$, $0 < s < 1$. For all $u \in W^{s,p}(\mathbb{R}^N)$, we have

$$u^\# \in W^{s,p}(\mathbb{R}^N) \text{ and } \|u^\#\|_{W^{s,p}} \leq \|u\|_{W^{s,p}}.$$

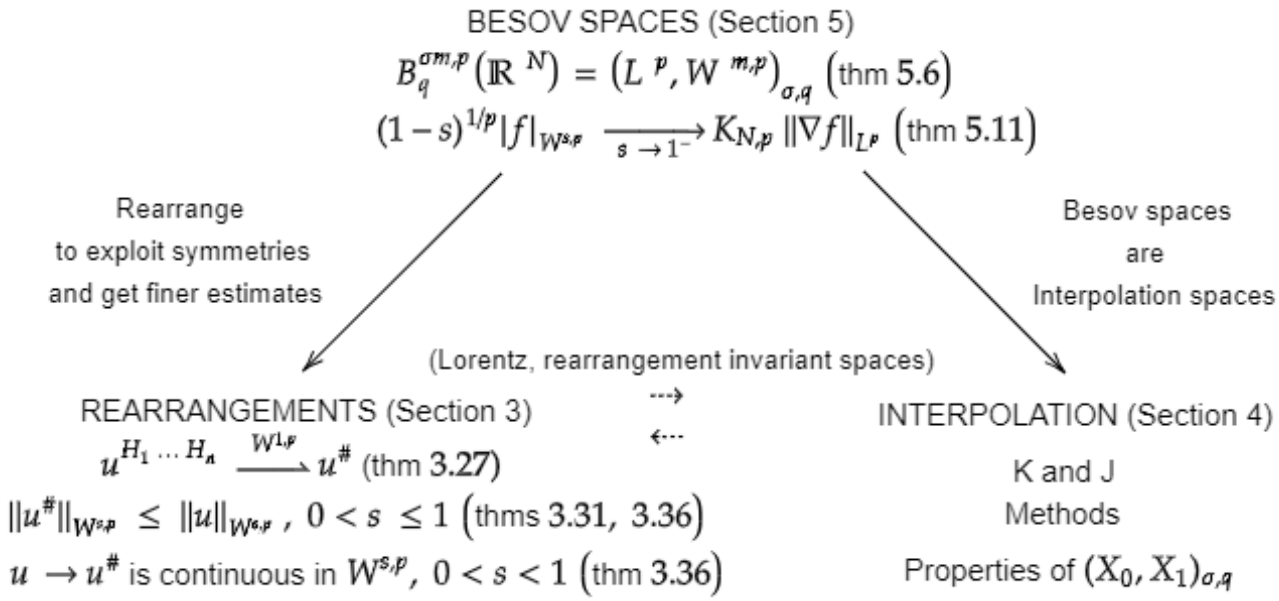
This results also holds for $W^{1,p}(\mathbb{R}^N)$. However, the fractional case is special, as the operation of rearrangement itself is continuous in $W^{s,p}$, and not in $W^{1,p}$. These results can be seen as consequences of Hardy-Littlewood-type inequalities on the spherically symmetric rearrangement and one can establish the latter by approximation through successive polarizations of a function (see subsections 3.4 and 3.5).

This illustrates how rearrangements, in general, allow to have a better control on estimates, as one chooses the function of same size that suits the problem best. Rearrangements often give the best idea of optimizers too: this is the case for Sobolev or Hardy-Littlewood-Sobolev inequalities, for example.

With this master thesis, we aim to study questions suggested by interactions between interpolation theory, Sobolev spaces, and Rearrangements.

For the sake of clarity and convenience, we chose to study rearrangements first; notions on these matters can be useful to put subsequent developments into perspective, if not absolutely necessary to grasp concepts and proofs. The order of the remaining parts is imposed. We need to take a step back to understand general interpolation, in order to apply this knowledge to the concrete example of Besov spaces.

Sections 3 and 5 are constructed around proofs, while section 4 consists only in the introduction and discussion of notions of interpolation theory. This overview is resumed in the following figure, where one can read the main theorems we will investigate under each topic.



Some notation and definitions: Throughout these notes

- N is the space dimension.
- For all $x \in \mathbb{R}^N$ $\|x\| := \left\{\sum_{i=1}^N x_i^2\right\}^{\frac{1}{2}}$.
- $B(x,r) := \{y \in \mathbb{R}^N,\ \|y-x\| < r\}$ for all $r > 0$, $x \in \mathbb{R}^N$.
- For μ an outer measure defined on the subsets of some R we denote $(R,\mu) := (R, \tau_\mu, \mu)$, where τ_μ is the complete (relatively to μ) σ-algebra of μ-measurable subsets of R.
- $|A|$ is the Lebesgue measure of a borel set $A \subset \mathbb{R}^N$.
- $\mathcal{H}^s$ is the s-dimensional Hausdorff measure.
- $\alpha_N := |B(0,1)|$, the volume of the unit ball in $\mathbb{R}^N$; $\beta_n := \mathcal{H}^{N-1}(\mathbb{S}^{N-1})$.
- In the particular case of dimension 1, we denote by m the Lebesgue measure.
- Given a measure space (R,μ), $M(R,\mu)$ is the set of μ-measurable functions $f : R \to [0,\infty]$, such that $f^{-1}(B) \in \tau_\mu$, for all $B \in \mathcal{B}([0,\infty])$.
- $M_0(R,\mu)$ is the set of almost everywhere (w.r.t. μ) finite functions.
- Given (R,μ) a measure space, we denote by $L^p(R,\mu)$ the Lebesgue space of classes of μ-measurable functions f such that $\|f\|_{L^p(R,\mu)} := (\int_R |f|^p d\mu)^{1/p} < \infty$.
- When we work on $\mathbb{R}^N$, we omit the reference to the Lebesgue measure or the Borel σ-algebra, as we only work on the measure space $(\mathbb{R}^N, \mathcal{B}(\mathbb{R}^N), |\cdot|)$. Same goes when we work on a Borel set $A \subset \mathbb{R}^N$ because we always refer to the measure space induced by $(\mathbb{R}^N, \mathcal{B}(\mathbb{R}^N), |\cdot|)$.

- In this vein, when working on functions defined on a Borel set $A \in \mathbb{R}^N$, we write about measurable functions, and measurable sets, as it is always with respect to the Borel σ-algebra.
  In this case, we write $L^p(A)$ to denote the Lebesgue space on A, with respect to the Lebesgue measure.

- In this case, when the domain of functions studied is clear, we sometimes omit any reference to it and write L^p, $W^{m,p}$, $B^{s,p}_q$ etc.

- Same goes for $M := M(\mathbb{R}^N, |\cdot|)$ and $M_0 = M_0(\mathbb{R}^N, |\cdot|)$.

- Given a multi-index $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_n)$ and a function f, we write $\partial^\alpha_D f = \partial^{\alpha_1}_{1,D} \ldots \partial^{\alpha_n}_{n,D} f$ with the partial derivatives being in the sense of distributions. The pointwise, traditional derivatives are denoted with a P instead of a D.
  These notations are only used when confusion can occur; most of the times, the way we differentiate is clear, and we just write ∂^α.

- Given an open set $\Omega \subset \mathbb{R}^N$ we define Sobolev spaces as (respectively "inhomogeneous" and "homogeneous" Sobolev spaces)

$$W^{m,p} := \{f \in L^p(\Omega);\ \partial^\alpha_D f \in L^p(\Omega), \forall \alpha \text{ s.t } |\alpha| \leq m\}$$

$$\dot{W}^{m,p} := \{f \in L^p_{loc}(\Omega);\ \partial^\alpha_D f \in L^p(\Omega), \forall \alpha \text{ s.t } |\alpha| = m\}.$$

- Given $f \in W^{m,p}(\Omega)$, we denote by $D^m f$ the vector formed by all partial derivatives of order m.

- We equip $W^{m,p}$ with the complete norm

$$\|f\|_{W^{m,p}} := \|f\|_{L^p} + \sum_{k=1}^{m} \big\| \|D^m f\| \big\|_{L^p}$$

  and $\dot{W}^{m,p}$ with the seminorm

$$|f|_{W^{m,p}} := \big\| \|D^m f\| \big\|_{L^p}.$$

- Finally, fractional Sobolev spaces are, for $0 < s < 1$,

$$W^{s,p}(\Omega) := \big\{ f \in L^p(\Omega),\ |f|^p_{W^{s,p}(\Omega)} := \int_{\Omega\times\Omega} \frac{|f(x) - f(y)|^p}{\|x-y\|^{sp+N}}\, dxdy < \infty \big\}$$

$$\dot{W}^{s,p}(\Omega) := \big\{ f \in L^p_{loc}(\Omega),\ |f|^p_{W^{s,p}(\Omega)} := \int_{\Omega\times\Omega} \frac{|f(x) - f(y)|^p}{\|x-y\|^{sp+N}}\, dxdy < \infty \big\}.$$

  We equip $W^{s,p}(\Omega)$ with the complete norm

$$\|f\|_{W^{s,p}(\Omega)} := \|f\|_{L^p} + |f|_{W^{s,p}}$$

  and $\dot{W}^{s,p}(\Omega)$ with the seminorm $|\cdot|_{W^{s,p}(\Omega)}$.

3 Rearrangements

Before studying particular rearrangements, we must define what it means to rearrange a function.

Considering real numbers $(a_i)_{1\leq i\leq m}$, a rearrangement of this family would be another family $(a_{\sigma(i)})_{1\leq i\leq m}$ with $\sigma \in S_m$. Observe that, in fact,

$$|\{1 \leq i \leq m\,;\; a_i > t\}| = |\{1 \leq i \leq m\,;\; b_i > t\}| \;\; \forall t \in \mathbb{R}$$
$$\iff \text{ there exists } \sigma \in S_m \text{ such that } (b_i) = (a_{\sigma(i)}).$$

This is the observation we are going to generalize to any measurable space.

Definition 3.1. Let (R, μ) a measure space and $f : R \to [0, \infty]$ be a μ measurable function. Consider the function $\mu_f : [0, \infty) \to [0, \infty]$ defined by $\mu_f(a) = \mu(\{x \in R\,;\; |f| > a\})$ for any $a \geq 0$. We call this function the "distribution function" of f.
We say that two functions $f \in M_0(R, \mu)$ and $g \in M_0(X, \nu)$ are equimeasurable if their distribution functions μ_f and ν_g are equal.
In this case, we also say that g is a rearrangement of f.

This section builds on questions on the spherically symmetric rearrangement. We prove that the spherically symmetric rearrangement $\#$ of a function in $W^{s,p}$ is in $W^{s,p}$, with precise estimates (thms 3.31 and 3.36), and that the operation of rearrangement itself is continuous in certain function spaces (thm 3.36). To do so we rely on Riesz-type inequalities for polarization (thm 3.22), that we prove using a pointwise approach, and on the approximation of $\#$ by successive polarizations in $W^{1,p}$ (thm 3.27).

3.1 Decreasing rearrangement

Arguably, the most natural way of organizing a family of real values is by following their size.

It is particularly useful to the analyst who knows that a monotone function is, compared to just any function, essentially trivial.

And, indeed, the decreasing rearrangement comes with a handful of properties that can drastically simplify our study of function spaces, if we use it carefully.

Definition 3.2. Let (R, μ) be a measure space and $f \in M_0(R, \mu)$. We can define its decreasing rearrangement by

$$f^*(t) := \inf\{\lambda \geq 0,\; \mu_f(\lambda) \leq t\},\; \forall t \geq 0.$$

f^* is indeed non-increasing. For some general properties of the rearrangement, see [1, Proposition 1.7]. Most of those properties come from the observation that $f^* = m_{\mu_f}$. Most importantly, f^* is indeed a rearrangement of f:

$$\mu_f(t) = m_{f^*}(t), \text{ for all } t > 0.$$

A useful consequence of the monotonicity of f^* is that we know where most of its mass is. On the other hand, one can transfer estimates from f^* to f as shows, for example, the Hardy-Littlewood inequality.

Theorem 3.3. If f and g belong to $M_0(R, \mu)$ then

$$\int_R |fg|\, d\mu \leq \int_0^\infty f^* g^*\, dx.$$

In some Banach spaces, there is a close relation between the sizes of f^* and f. Here is a simple illustration of this fact.

Proposition 3.4.

$$\|f\|_{L^p(R)}^p = p\int_0^\infty \lambda^{p-1}\mu_f(\lambda)\,d\lambda = p\int_0^\infty \lambda^{p-1}m_{f^*}(\lambda)\,d\lambda = \|f^*\|_{L^p(\mathbb{R}^+)}^p$$

More generally, it holds that one can determine, in **rearrangement invariant function spaces**, the norm of a function by calculating a (possibly different) norm of its decreasing rearrangement. This is the content of the very important **Luxembourg representation theorem**.

Finally, we mention that a systematic account of rearrangement invariant Banach function spaces can be found in [1, Chapter 2].

3.2 Two nice ways to rearrange a function in $\mathbb{R}^N$

In Euclidean spaces, one may rearrange the values of a function in order to exploit our most basic geometric intuitions on symmetry, volume, surfaces etc. combined with the analytic approach above.

With this in mind, we feel like we are losing a lot of information taking the one dimensional decreasing rearrangement of a function of the space variable. However, just a minor modification of the above construction allows to recover all valuable features of $*$ in a more geometric setting. A complete account of the matters addressed in this section may be found in [6, Chapter 15] and [7, Chapters 4,10]

Definition 3.5. (Spherically symmetric rearrangement) Let $f : \mathbb{R}^N \to \mathbb{R}$ be a measurable function. We define its spherically symmetric rearrangement $f^\#$ by

$$f^\#(x) := f^*(\alpha_N\|x\|^N),\ \forall x \in \mathbb{R}^N.$$

We also define the spherically symmetric rearrangement of a measurable set E by

$$E^\# := B\left(0, \left\{\frac{|E|}{\alpha_N}\right\}^{1/N}\right)$$

It is the open ball centered on 0 with the same Lebesgue measure as E. One can easily prove, using the properties of the one dimensional rearrangement, that $f^\#$ is l.s.c. Using a change of variables, one may apply Proposition 3.4 to find that

$$\int_{\mathbb{R}^N} |f(x)|^p\,dx = \int_{\mathbb{R}^N} |f^\#(x)|^p\,dx \text{ and } \operatorname{ess\,sup}_{x\in\mathbb{R}^N}|f| = \operatorname{ess\,sup}_{x\in\mathbb{R}^N}|f^\#|$$

for all $1 \le p < \infty$.

For other properties similar to those of f^*, see [6, Proposition 15.6].

The spherically symmetric rearrangement of functions and sets are "equivalent" notions (see Remark 3.7 below), as suggests

Observation 3.6. $\{x \in \mathbb{R}^N, u^\#(x) > t\} = \{x \in \mathbb{R}^N, |u(x)| > t\}^\#$, for all $t > 0$.

Remark 3.7. Here is a more geometrical approach to the definition of $f^\#$. We first define what the rearrangement of a set $E^\#$ is, and then rearrange the values of f in $f^\#$ in order to have $\{f^\# > t\} = \{|f| > t\}^\#$ through the layer-cake representation:

$$f^\#(x) := \int_0^\infty \mathbb{1}_{\{|f|>t\}^\#}(x,t)dt.$$

The definitions coincide as Observation 3.6 shows.

In view of the developments below (see, for example, the proof of Theorem 3.22), we ought to know how $\#$ behaves with respect to some standard approximation procedures. Proofs of the following two propositions can be found in [7, Chapter 4]. One reads in these pages that the functions must be "vanishing at infinity" to be able to conclude to the results below. In fact, the proofs in [7] work without this assumption.

Proposition 3.8. Let (u_n) be a sequence of nonnegative measurable functions such that $u_n \xrightarrow[n\to\infty]{} u$ in measure, for some measurable function $u : \mathbb{R}^N \to \mathbb{R}^+$. Then $u_n^\#(x) \xrightarrow[n\to\infty]{} u^\#(x)$ almost everywhere.

Proposition 3.9. Let (u_n) be a sequence of nonnegative measurable functions defined on $\mathbb{R}^N$, such that $u_n \leq u_{n+1}$ for all n. Set $u(x) := \lim_n u(x)$ for all $x \in \mathbb{R}^N$. Then $u_n^\# \xrightarrow[n\to\infty]{} u^\#$ for all $x \in \mathbb{R}^N$.

We next introduce another geometric rearrangement that is easier to manipulate than $\#$: the rearrangement by polarization. This simplicity has a price: polarizing a function f gives an object that is less symmetric than $f^\#$; on the other hand, as it only consists in a slight modification of f, it allows calculations that are much more difficult to perform when dealing directly with $f^\#$.

The interest of the polarization stands in the fact that it is a simple operation allowing to recover the spherical symmetrization via a limiting process, see section 3.4.

Definition 3.10. Consider an affine hyperplane H, ν one its normal vectors, σ the reflection associated to this hyperplane, and $P_{\vec{H}}$ the orthogonal projection on the linear direction $\vec{H}$ of H. There exists $a \in \mathbb{R}$ such that

$$H = \{x \in \mathbb{R}^n, \nu \cdot x = a\},\ \sigma(x) = P_{\vec{H}}(x) + (2a - \nu \cdot x)\nu.$$

If H does not contain the origin, we take ν such that $a > 0$.

In that case, we denote $H^+ := \{\nu \cdot x < a\}$ and $H^- := \{\nu \cdot x > a\}$. Thus H^+ is the connected component of $\mathbb{R}^N \setminus H$ containing 0. If H is linear, denote arbitrarily H^+ and H^- the two components of $\mathbb{R}^N \setminus H$.

We are, finally, in position to define the polarization u^H of u by

$$u^H(x) = \begin{cases} \max\{u(x), u(\sigma(x))\} & \text{if } x \in H^+ \\ \min\{u(x), u(\sigma(x))\} & \text{if } x \in H^- \\ u(x) & \text{if } x \in H \end{cases}.$$

In the following, we stick to these notations. Each time we are considering the polarization u^H of a function u, we denote by σ the reflection associated to the hyperplane H.

Proposition 3.11. u^H is indeed a rearrangement of u. More specifically, it holds that

$$|\{x \in \mathbb{R}^N \,;\, u(x) > t\}| = |\{x \in \mathbb{R}^N \,;\, u^H(x) > t\}|, \ \forall t \in \mathbb{R}.$$

The only thing we are doing when polarizing a function is to exchange values when needed to obtain a function that is decreasing across H.

Therefore, one can estimate the modulus of continuity and the support of a polarization u^H of a function u, by corresponding estimates on u. One finds a richer review of similar properties in [7, Chapter 4].

Definition 3.12. Let (X, d_X) and (Y, d_Y) be two metric spaces. We define the modulus of continuity $\omega_f(\cdot)$ of a function $f : X \to Y$ by

$$\omega_f(t) := \sup\{d_Y(f(x), f(y)) \,;\, x, y \in X \text{ such that } d_X(x, y) \leq t\}, \text{ for all } t > 0.$$

Proposition 3.13. Let u be a measurable function, and H an affine hyperplane. Then, we have that $\omega_{u^H} \leq \omega_u$.

The above result essentially relies on the following observations.

Lemma 3.14. Let H be an affine hyperplane and let σ be the corresponding reflection. Then

- $\|x - y\| = \|\sigma(x) - \sigma(y)\|, \ \forall x, y \in \mathbb{R}^N$.
- $\|x - \sigma(y)\| = \|\sigma(x) - y\| \leq \|x - y\|$, for all $x \in H^+, y \in H^-$.

This simple observation will be crucial in many developments below.
We have a direct application of this in the next important result.

Proposition 3.15. Let u be a measurable function, and H an affine hyperplane. Then

$$\text{diam}\{x \in \mathbb{R}^N \,;\, u^H(x) > t\} \leq \text{diam}\{x \in \mathbb{R}^N \,;\, u(x) > t\}, \ \forall t \in \mathbb{R}.$$

We can even go further and differentiate.

Theorem 3.16. ([7, p.393]) Let u be in $\dot{W}^{1,p}(\mathbb{R}^N)$ and let H be an affine hyperplane. Then u^H is in $W^{1,p}(\mathbb{R}^N)$ and

$$\text{for a.e } x \in \mathbb{R}^N, \ \nabla u^H(x) = \begin{cases} \nabla u(x) & \text{if } x \in H^+ \text{ and } u(x) > u(\sigma(x)) \\ -\nabla u(\sigma(x)) & \text{if } x \in H^+ \text{ and } u(x) < u(\sigma(x)) \\ \nabla u(x) & \text{if } x \in H^- \text{ and } u(x) < u(\sigma(x)) \\ -\nabla u(\sigma(x)) & \text{if } x \in H^- \text{and } u(x) > u(\sigma(x)) \\ \nabla u(x) & \text{if } u(\sigma(x)) = u(x) \end{cases}.$$

Combining this result and definition 3.10, one finds there exists $\Phi_{H,u}$ bijective and measure-preserving, such that $|\nabla u^H| = |\nabla u| \circ \Phi_{H,u}$, $u^H = u \circ \Phi_{H,u}$ almost everywhere.

Explicitly,

$$\Phi_{H,u}(x) = \begin{cases} x & \text{if } x \in H^+ \cap \{u > u \circ \sigma\} \sqcup H^- \cap \{u < u \circ \sigma\} \sqcup \{u = u \circ \sigma\} \\ \sigma(x) & \text{if } x \in H^+ \cap \{u < u \circ \sigma\} \sqcup H^- \cap \{u > u \circ \sigma\} \end{cases}$$

Therefore, $\|\nabla u^H\|_{L^p} = \|\nabla u\|_{L^p}$ and $\|u^H\|_{L^p} = \|u\|_{L^p}$.

3.3 Polarization: Hardy-Littlewood type inequalities and Sobolev norms

As the spherically symmetric rearrangement, the polarization verifies a Hardy-Littlewood inequality (Theorem 3.3).

This section is a survey of a few inequalities of this type that one can establish directly using pointwise inequalities based on convexity.

The only proof we present is the one of an inequality particularly relevant in the study of Sobolev spaces. However, all the results in this section can be proven independently, using the same straight-forward approach as in the proof of Theorem 3.20.

This approach is different from the one used by Leoni in [6, 117-123], which consists in proving Theorem 3.20 as a corollary of Theorem 3.18, and Theorem 3.19 as a corollary of Theorem 3.17. We chose to present it because it seemed more natural for us to proceed that way. Additionally, this approach allows to write proofs that use weaker assumptions than in [7]. For example, in Theorem 3.20, there is no need for g to be vanishing at 0, and it applies to any measurable functions u and v.

Theorem 3.17. (Hardy-Littlewood) Let H be an affine hyperplane and u, v be measurable functions such that the integrals on $\mathbb{R}^N$ of uv and $u^H v^H$ are well-defined. Then

$$\int_{\mathbb{R}^N} uv\,dx \leq \int_{\mathbb{R}^N} u^H v^H\,dx.$$

Moreover, if one of the integrals is finite, then equality holds if and only if

$$\Big(u(x) - u(\sigma(x))\Big)\Big(v(x) - v(\sigma(x))\Big) \geq 0 \text{ a.e.}$$

Theorem 3.18. Let H be an affine hyperplane and g be a nonnegative convex function. For all u, v measurable,

$$\int_{\mathbb{R}^N} g(v^H(x) - u^H(x))\,dx \leq \int_{\mathbb{R}^N} g(v(x) - u(x))\,dx.$$

This is particularly important in the case where $g(x) = |x|^p$, which asserts that

$$\int_{\mathbb{R}^N} \Big|v^H(x) - u^H(x)\Big|^p dx \leq \int_{\mathbb{R}^N} \Big|v(x) - u(x)\Big|^p dx.$$

Consequently, $L^p \ni u \to u^H$ is continuous from L^p to L^p.

We now turn to what we could call Riesz rearrangement inequalities.

Theorem 3.19. Let $k : \mathbb{R}^+ \to \mathbb{R}^+$ be a non-increasing measurable function, H an affine hyperplane and $u, v : \mathbb{R}^N \to \mathbb{R}^+$ measurable. Then

$$\int_{\mathbb{R}^N \times \mathbb{R}^N} u(x)v(y)k(\|x-y\|)\,dxdy \leq \int_{\mathbb{R}^N \times \mathbb{R}^N} u^H(x)v^H(y)k(\|x-y\|)\,dxdy.$$

Theorem 3.20. Let $u, v : \mathbb{R}^N \to \mathbb{R}$ be measurable functions, $g : \mathbb{R} \to \mathbb{R}^+$ convex, and $k : \mathbb{R}^N \to \mathbb{R}^+$ non-increasing. Then

$$\int_{\mathbb{R}^N}\int_{\mathbb{R}^N} g(v^H(y) - u^H(x))k(\|x-y\|)\,dxdy \leq \int_{\mathbb{R}^N}\int_{\mathbb{R}^N} g(v(y) - u(x))k(\|x-y\|)\,dxdy$$

To prove Theorem 3.20, we aim for the most straight-forward way of showing an inequality between integrals, by showing that the inequality holds for the functions themselves. As calculations will reveal, the symmetries in the integrands, induced by the presence of $k(\|x-y\|)$ and the monotonicity of k, will allow us to reduce the desired pointwise inequality to a four point inequality for convex functions.

Observation 3.21. Consider four distinct real numbers x, y, z, t, with $x \leq y$. We have

$$\begin{cases} x+y=z+t \\ z,t \in [x,y] \end{cases} \implies g(x)+g(y) \geq g(z)+g(t), \text{ for all } g \text{ convex}$$

The above implication is a straightforward application of Jensen's inequality. Let us proceed to the proof of Theorem 3.20.

Proof of Theorem 3.20. We have

$$\begin{aligned}
&\int_{\mathbb{R}^N \times \mathbb{R}^N} g(v^H(y)-u^H(x))k(\|x-y\|)\,dxdy \\
&= \int_{H^+\times H^+} g(v^H(y)-u^H(x))k(\|x-y\|)\,dxdy + \int_{H^+\times H^-} g(v^H(y)-u^H(x))k(\|x-y\|)\,dxdy \\
&+ \int_{H^-\times H^+} g(v^H(y)-u^H(x))k(\|x-y\|)\,dxdy + \int_{H^-\times H^-} g(v^H(y)-u^H(x))k(\|x-y\|)\,dxdy \\
&= \int_{H^+\times H^+} \Big\{ g(v^H(y)-u^H(x))k(\|x-y\|) + g(v^H(\sigma(y))-u^H(x))k(\|x-\sigma(y)\|) \\
&+ g(v^H(y)-u^H(\sigma(x)))k(\|\sigma(x)-y\|) + g(v^H(\sigma(y))-u^H(\sigma(x))k(\|x-y\|) \Big\}\,dxdy \\
&:= \int_{H^+\times H^+} G^H(x,y)\,dxdy.
\end{aligned}$$

Similarly, we have

$$\begin{aligned}
&\int_{\mathbb{R}^N}\int_{\mathbb{R}^N} g(v(y)-u(x))k(\|x-y\|)\,dxdy \\
&= \int_{H^+\times H^+} \Big\{ g(v(y)-u(x))k(\|x-y\|) + g(v(\sigma(y))-u(x))k(\|x-\sigma(y)\|) \\
&+ g(v(y)-u(\sigma(x)))k(\|\sigma(x)-y\|) + g(v(\sigma(y))-u(\sigma(x))k(\|x-y\|) \Big\}\,dxdy \\
&:= \int_{H^+\times H^+} G(x,y)\,dxdy.
\end{aligned}$$

The desired inequality

$$\int_{H^+\times H^+} G^H(x,y)\,dxdy \leq \int_{H^+\times H^+} G(x,y)dxdy$$

follows from the following

Claim : $G^H(x,y) \leq G(x,y),\ \forall (x,y) \in H^+ \times H^+$.

Indeed, considering $\delta(x,y) := G^H(x,y) - G(x,y)$:

- If $v(y) \geq v(\sigma(y))$ and $u(x) \geq u(\sigma(x))$, we are in the case where :

$$u^H(x) = u(x),\ u^H(\sigma(x)) = u(\sigma(x)),\ v^H(y) = v(y),\ v^H(\sigma(y)) = v(\sigma(y))$$
$$\text{and } G(x,y) = G^H(x,y).$$

- The symmetric case where $v(y) < v(\sigma(y))$ and $u(x) < u(\sigma(x))$ is similar to the first one.

- If $u(x) \geq u(\sigma(x))$ and $v(y) < v(\sigma(y))$, we have $u^H(x) = u(x)$, $u^H(\sigma(x)) = u(\sigma(x))$, $v^H(y) = v(\sigma(y))$, $v^H(\sigma(y)) = v(y)$, and

$$\begin{aligned}\delta(x,y) = &\Big\{g(v(\sigma(y)) - u(x)) + g(v(y) - u(\sigma(x)))\Big\}k(\|x-y\|)\\ &+\Big\{g(v(\sigma(y)) - u(\sigma(x))) + g(v(y) - u(x))\Big\}k(\|\sigma(x)-y\|)\\ &-\Big\{g(v(y) - u(x)) + g(v(\sigma(y)) - u(\sigma(x)))\Big\}k(\|x-y\|)\\ &+\Big\{g(v(\sigma(y)) - u(x))) + g(v(y) - u(\sigma(x)))\Big\}k(\|\sigma(x)-y\|)\end{aligned}$$

  We find that

$$\delta(x,y) = \Big[k(\|x-y\|) - k(\|\sigma(x)-y\|)\Big]$$
$$\Big\{g(v(\sigma(y) - u(x)) + g(v(y) - u(\sigma(x))) - \Big(g(v(y) - u(x)) + g(u(\sigma(y)) - u(\sigma(x)))\Big)\Big\}$$

  We next note that $k(\|x-y\|) - k(\|\sigma(x)-y\|) \geq 0$ (since $\|x-y\| \leq \|\sigma(x)-y\|$ for $x, y \in H^+$). Combining this with Observation 3.21, applied with

$$x := v(y) - u(x)\ ,\ y := v(\sigma(y)) - u(\sigma(x))\ ,\ z := v(\sigma(y)) - u(x)\ ,\ t := v(y) - u(\sigma(x)),$$

  we find that $\delta(x,y) \leq 0$.

- The last case where $u(x) < u(\sigma(x))$ and $v(y) \geq v(\sigma(y))$ is similar to the previous one.

In conclusion, in all cases, $\delta(x,y) = G^H(x,y) - G(x,y) \leq 0$ and $G^H(x,y) \leq G(x,y)$ for all $x, y \in H^+$. As we observed above, this inequality integrated on $H^+ \times H^+$ gives the desired estimate. □

3.4 Approximating the spherically symmetric rearrangement with iterated polarizations

In this section, we will see how an appropriate iterative process involving successive polarizations of a given function u yields, in the limit, its spherical symmetrization $u^\sharp$. A major consequence of this is a systematic proof of (difficult) inequalities involving $u^\sharp$ starting from the (easy) inequalities involving u^H mentioned in the previous section.

We will illustrate this systematic procedure in the proof of Theorem 3.22 below.

Theorem 3.22. Let $1 < p \leq \infty$. Let $u, v : \mathbb{R}^N \to \mathbb{R}$ be in $L^p(\mathbb{R}^N)$ and $k : (0,\infty) \to \mathbb{R}^+$ be non-increasing. Then

$$\int_{\mathbb{R}^N}\int_{\mathbb{R}^N} |v^\#(y) - u^\#(x)|^p k(\|x-y\|)\,dxdy \leq \int_{\mathbb{R}^N}\int_{\mathbb{R}^N} |v(y) - u(x)|^p k(\|x-y\|)\,dxdy.$$

Let $m \geq 2$. Given m hyperplanes $H_1, \dots, H_m$, and f a measurable function, we define inductively $f^{H_1 \dots H_m} := (f^{H_1 \dots H_{m-1}})^{H_m}$.

We denote, for u, v measurable functions,

$$P_u := \{u^{H_1 \dots H_N},\ n \in \mathbb{N}^*, H_1, \dots, H_n \text{ hyperplanes}\}$$

$$P_{u,v} := \{(u^{H_1 \dots H_N}, v^{H_1 \dots H_N}),\ n \in \mathbb{N}^*, H_1, \dots, H_n \text{ hyperplanes}\}.$$

In order to prove Theorem 3.22, we will rely on the possibility of approximating the spherical rearrangement of a function u by functions in P_u, as announced at the beginning of this section. However, we will need the slightly stronger

Theorem 3.23. Let $u, v \in C_c(\mathbb{R}^N, \mathbb{R}^+)$. There exists some $R > 0$, and some sequence $(u_n, v_n) \subset P_{u,v}$ such that

$$u_n \underset{n\to\infty}{\longrightarrow} u^\# \text{ and } v_n \underset{n\to\infty}{\longrightarrow} v^\# \text{ uniformly},$$

with supp $u_n \subset B(0, R)$, supp $v_n \subset B(0, R)$.

One may find a proof of Theorem 3.23 in [7, p.137].

Proof of Theorem 3.22. Note that one only needs to prove this result for nonnegative functions. Indeed, for all functions $u, v : \mathbb{R}^N \to \mathbb{R}$, one has that

$$||v|(y) - |u|(x)| \leq |v(y) - u(x)| \text{ for all } x, y \in \mathbb{R}^N \text{ and } u^\# = (|u|)^\#, v^\# = (|v|)^\#.$$

Let us proceed to the proof in that case.

Since Theorem 3.4 only applies to $C_c(\mathbb{R}^N)$ functions, one needs to proceed in several steps to get to the full result.

1. One transforms a result about polarizations (Theorem 3.20) into a result about spherically symmetric rearrangements (Theorem 3.22) of $C_c(\mathbb{R}^N)$ functions, using Theorem 3.4.

2. Knowing that Theorem 3.27 holds in this case, one generalizes this result, from $C_c(\mathbb{R}^N, \mathbb{R}^+)$ to all $L^p(\mathbb{R}^N)$ functions. This can be done by a density argument. However, one needs to be in position to apply dominated convergence (or at least something stronger than Fatou's lemma) on the right hand side of the inequality in Theorem 3.22.

   In this perspective, one may assume, in a first place, that k is bounded and with compact support to show that the theorem holds in this case.

   We take these assumptions on k to assure that a function like

   $$(x, y) \to |v(y) - u(x)|^p k(\|x - y\|)$$

   is indeed integrable (see Step 2 of the proof below).

3. It is then easy to generalize to Theorem 3.22, approximating any k by an increasing sequence of bounded, non-increasing functions with compact support, and using monotone convergence.

1. **We show that Theorem 3.22 holds for u in $C_c(\mathbb{R}^N)$ and nonnegative.** We approximate $(u^\#, v^\#)$ with a sequence (u_n, v_n) in $P_{u,v}$ (by Theorem 3.4). Reiterating Theorem 3.20 along (u_n, v_n) shows that, for all $n \in \mathbb{N}$,

$$\int_{\mathbb{R}^N}\int_{\mathbb{R}^N} |v_n(y) - u_n(x)|^p k(\|x - y\|)\, dxdy \leq \int_{\mathbb{R}^N}\int_{\mathbb{R}^N} |v(y) - u(x)|^p k(\|x - y\|)\, dxdy.$$

Since $u_n \to u^\#$ and $v_n \to v^\#$ pointwise, applying Fatou's lemma allows to conclude that Theorem 3.22 holds in the case where $u, v \in C_c(\mathbb{R}^N, \mathbb{R}^+)$.

2. Suppose that k is bounded and with support in an interval $[0, M]$, for some $M > 0$. Let $u, v : \mathbb{R}^n \to \mathbb{R}^+$ be functions in $L^p(\mathbb{R}^N)$.

We can approximate u and v by sequences (u_n) and (v_n) of $C_c(\mathbb{R}^N, \mathbb{R}^+)$ functions in $L^p(\mathbb{R}^N)$. Moreover, those sequences can be constructed so as to verify:
1) $u_n \to u$ and $v_n \to v$ a.e.
2) There exists $\mathcal{U}$ and $\mathcal{V}$ in L^1 such that $|u_n|^p \leq \mathcal{U}$ and $|v_n|^p \leq \mathcal{V}$ a.e.

An application of the first step of the proof gives that, for all $n \in \mathbb{N}$,

$$(R_n) \quad \int_{\mathbb{R}^N}\int_{\mathbb{R}^N} |v_n^\#(y) - u_n^\#(x)|^p k(\|x-y\|)\, dxdy \leq \int_{\mathbb{R}^N}\int_{\mathbb{R}^N} |v_n(y) - u_n(x)|^p k(\|x-y\|)\, dxdy.$$

The right hand side of (R_n) converges to $\int_{\mathbb{R}^N\times\mathbb{R}^N} |v(y)-u(x)|^p k(\|x-y\|)\, dxdy$ by dominated convergence.
(Domination) For all $n \in \mathbb{N}$,

$$|v_n(y)-u_n(x)|^p k(\|x-y\|) \leq 2^{p-1}\{|v_n(y)|^p + |u_n(x)|^p\} k(\|x-y\|) \leq 2^{p-1}(\mathcal{V}+\mathcal{U}) k(\|x-y\|) \text{ a.e.}$$

This domination verifies the assumptions of the dominated convergence as the function $(\mathcal{V}(y) + \mathcal{U}(x))k(\|x-y\|)$ is $L^1(\mathbb{R}^N \times \mathbb{R}^N)$.

$$\begin{aligned}\int_{\mathbb{R}^N\times\mathbb{R}^N} (\mathcal{V}+\mathcal{U}) k(\|x-y\|)\, dxdy &\leq \|k\|_\infty \int_{\|x-y\|<M} (\mathcal{V}(y)+\mathcal{U}(x)) dxdy \\ &\leq \|k\|_\infty \left\{ \int_{\mathbb{R}^N} \mathcal{V}(y) \left(\int_{B(y,M)} dx \right) dy + \int_{\mathbb{R}^N} \mathcal{U}(x) \left(\int_{B(x,M)} dy \right) dx \right\} < \infty.\end{aligned}$$

(Poinwise convergence) $|v_n(y) - u_n(x)|^p k(\|x-y\|) \to |v(y) - u(x)|^p k(\|x-y\|)$ almost everywhere, since $u_n \to u$ and $v_n \to v$ almost everywhere.

The left hand side, at the limit verifies

$$\int_{\mathbb{R}^N\times\mathbb{R}^N} |v^\#(y)-u^\#(x)|^p k(\|x-y\|)\, dxdy \leq \liminf_n \int_{\mathbb{R}^N\times\mathbb{R}^N} |v_n^\#(y)-u_n^\#(x)|^p k(\|x-y\|)\, dxdy.$$

This comes from an application of Fatou's lemma since $u_n^\# \to u^\#$ and $v_n^\# \to v^\#$ almost everywhere by Proposition 3.8.
Combining our observations on the left and right hand sides of (R_n), one can pass to the limit in (R_n) and find

$$\begin{aligned}&\int_{\mathbb{R}^N\times\mathbb{R}^N} |v^\#(y) - u^\#(x)|^p k(\|x-y\|)\, dxdy \\ &\leq \liminf_n \int_{\mathbb{R}^N\times\mathbb{R}^N} |v_n^\#(y) - u_n^\#(x)|^p k(\|x-y\|)\, dxdy \\ &\leq \lim_n \int_{\mathbb{R}^N}\int_{\mathbb{R}^N} |v_n(y) - u_n(x)|^p k(\|x-y\|)\, dxdy \\ &= \int_{\mathbb{R}^N\times\mathbb{R}^N} |v(y) - u(x)|^p k(\|x-y\|)\, dxdy.\end{aligned}$$

3. Finally, consider a non-increasing function $k : (0,\infty) \to \mathbb{R}^+$ and two functions $u, v : \mathbb{R}^N \to \mathbb{R}^+$ in $L^p(\mathbb{R}^N)$. One can approximate k by an increasing sequence of functions (k_n) that verify the assumptions in step 2.
Set $k_n := \mathbb{1}_{[0,n]} \min\{k, n\}$. The functions k_n verify the assumptions of step 2. Therefore, for all $n \in \mathbb{N}$,

$$\int_{\mathbb{R}^N}\int_{\mathbb{R}^N} |v^\#(y) - u^\#(x)|^p k_n(\|x-y\|)\,dxdy \le \int_{\mathbb{R}^N}\int_{\mathbb{R}^N} |v(y) - u(x)|^p k_n(\|x-y\|)\,dxdy.$$

Since the sequence (k_n) is increasing and converging to k pointwise, one may apply monotone convergence on both sides of the previous inequality and conclude that

$$\int_{\mathbb{R}^N}\int_{\mathbb{R}^N} |v^\#(y) - u^\#(x)|^p k(\|x-y\|)\,dxdy \le \int_{\mathbb{R}^N}\int_{\mathbb{R}^N} |v(y) - u(x)|^p k(\|x-y\|)\,dxdy. \quad \square$$

The same kind of proof allows to transfer Theorem 3.18 to spherical rearrangements.

Theorem 3.24. Let $1 \le p < \infty$, and $u, v : \mathbb{R}^N \to \mathbb{R}$ be functions in $L^p(\mathbb{R}^N)$. We have

$$\int_{\mathbb{R}^N} |v^\#(x) - u^\#(x)|^p\,dx \le \int_{\mathbb{R}^N} |v(x) - u(x)|^p\,dx.$$

Consequently, $L^p \ni u \to u^\# \in L^p$ is continuous.

Instead of proving Theorem 3.24 using the same approach as in the proof of Theorem 3.22, let us show that this result is a direct consequence of Theorem 3.22.

Proof of Theorem 3.24. We proceed so as to formalize one's intuition that Theorem 3.24 only consists in an application of Theorem 3.22 to $k = \delta_0$.

Let $u, v : \mathbb{R}^N \to \mathbb{R}$ be functions in L^p. For all $\delta > 0$, one can apply Theorem 3.22 to $k(t) = \mathbb{1}_{[0,\delta)}$, and find that

$$\int_{\mathbb{R}^N}\int_{\mathbb{R}^N} \left|v^\#(y) - u^\#(x)\right|^p \mathbb{1}_{[0,\delta)}(\|x-y\|)\,dxdy \le \int_{\mathbb{R}^N}\int_{\mathbb{R}^N} \left|v^\#(y) - u^\#(x)\right|^p \mathbb{1}_{[0,\delta)}(\|x-y\|)\,dxdy.$$

We may rewrite, with a change of variables, this inequality as

$$\int_{\mathbb{R}^N} \mathbb{1}_{[0,\delta)}(\|h\|) \int_{\mathbb{R}^N} \left|v^\#(x+h) - u^\#(x)\right|^p dx\,dh \le \int_{\mathbb{R}^N} \mathbb{1}_{[0,\delta)}(\|h\|) \int_{\mathbb{R}^N} \left|v^\#(x+h) - u^\#(x)\right|^p dx\,dh.$$

The change of variable $h = \delta z$ allows us to write

$$\delta^N \int_{\mathbb{R}^N} \mathbb{1}_{[0,1)}(\|z\|) \int_{\mathbb{R}^N} \left|v^\#(x+\delta z) - u^\#(x)\right|^p dx\,dz \le \delta^N \int_{\mathbb{R}^N} \mathbb{1}_{[0,1)}(\|z\|) \int_{\mathbb{R}^N} \left|v^\#(x+\delta z) - u^\#(x)\right|^p dx\,dz.$$

Therefore, we have

$$\int_{\mathbb{R}^N} \mathbb{1}_{[0,1)}(\|z\|)\left\|v^{\#}(\cdot+\delta z)-u^{\#}(\cdot)\right\|^p_{L^p(\mathbb{R}^N)}\,dz \\ \leq \int_{\mathbb{R}^N} \mathbb{1}_{[0,1)}(\|z\|)\|v(\cdot+\delta z)-u(\cdot)\|^p_{L^p(\mathbb{R}^N)}\,dz. \quad (3.1)$$

We aim to compute the limits of both sides of (3.1) as $\delta \to 0^+$, with an application of dominated convergence.

We need to make an observation before proceeding. We know that $u^{\#}, v^{\#}$ are in $L^p(\mathbb{R}^N)$, see Definition 3.5. On the other hand, we have that

$$\|f(\cdot+h)-f(\cdot)\|_{L^p(\mathbb{R}^N)} \xrightarrow[\|h\|\to 0]{} 0, \text{ for all } f \in L^p(\mathbb{R}^N).$$

This implies that, for all $z \in \mathbb{R}^N$,

$$\|v(\cdot+\delta z)-u(\cdot)\|^p_{L^p(\mathbb{R}^N)} \xrightarrow[\delta\to 0^+]{} \|v(\cdot)-u(\cdot)\|^p_{L^p(\mathbb{R}^N)}, \quad (3.2)$$

$$\left\|v^{\#}(\cdot+\delta z)-u(\cdot)\right\|^p_{L^p(\mathbb{R}^N)} \xrightarrow[\delta\to 0^+]{} \left\|v^{\#}(\cdot)-u(\cdot)\right\|^p_{L^p(\mathbb{R}^N)}. \quad (3.3)$$

We are in position to apply dominated convergence to (3.1). Let us consider the left hand side. (3.3) gives the desired pointwise convergence of the integrands. As for a domination, one finds that for all $\delta > 0$, for all $z \in \mathbb{R}^N$,

$$\mathbb{1}_{[0,1)}(\|z\|)\left\|v^{\#}(\cdot+\delta z)-u^{\#}(\cdot)\right\|^p_{L^p(\mathbb{R}^N)} \leq 2^{p-1}\mathbb{1}_{[0,1)}(\|z\|)\left\{\left\|v^{\#}\right\|_{L^p(\mathbb{R}^N)}+\left\|u^{\#}\right\|_{L^p(\mathbb{R}^N)}\right\}.$$

$2^{p-1}\mathbb{1}_{[0,1)}(\cdot)\{\left\|v^{\#}\right\|_{L^p(\mathbb{R}^N)}+\left\|u^{\#}\right\|_{L^p(\mathbb{R}^N)}\}$ is in $L^1(\mathbb{R}^N)$. Therefore, one may apply dominated convergence to find that, as $\delta \to 0^+$, the right hand side of (3.1) converges to

$$\int_{R^N} \mathbb{1}_{[0,1)}(\|z\|)\left\|v^{\#}-u^{\#}\right\|^p_{L^p(\mathbb{R}^N)}\,dz = \alpha_N\left\|v^{\#}-u^{\#}\right\|^p_{L^p(\mathbb{R}^N)}.$$

The exact same argument shows that the right hand side of (3.1) converges to $\alpha_N\|v-u\|^p_{L^p(\mathbb{R}^N)}$. Therefore, passing to the limit $\delta \to 0^+$ in (3.1), and cancelling α_N out, gives the desired estimate. □

One may proceed as in the proof of Theorem 3.22 to prove the following result, see [7, p.141].

Theorem 3.25. Let $k : (0,\infty) \to \mathbb{R}^+$ be a non-increasing measurable function, H an affine hyperplane and $u, v : \mathbb{R}^N \to \mathbb{R}^+$ measurable functions. Then

$$\int_{\mathbb{R}^N\times\mathbb{R}^N} u(x)v(y)k(\|x-y\|)\,dxdy \leq \int_{\mathbb{R}^N\times\mathbb{R}^N} u^{\#}(x)v^{\#}(y)k(\|x-y\|)\,dxdy.$$

An important approximation result is obtained by combining Theorem 3.23 with the continuity of $\#$ and of polarization.

Theorem 3.26. Let $u \in L^p$ be nonnegative. Then there exists a sequence $(u_n) \subset P_u$ such that $u_n \to u^{\#}$ in L^p.

Proof of Theorem 3.26. Let $\epsilon > 0$.
Let $\phi \in C_c^\infty$, be nonnegative such that $\|u-\phi\|_p \leq \epsilon$.
Since ϕ verifies the assumptions of Theorem 3.23, there exists $n \in \mathbb{N}^*$ and hyperplanes $H_1, \ldots, H_n$

such that $\|\phi^{H_1\dots H_n} - \phi^{\#}\|_p \leq \epsilon$ (we have uniform convergence to $u^{\#}$ of a sequence of polarizations with supports in a fixed ball). Therefore, $u^{H_1\dots H_n}$ is really close to $u^{\#}$ since:

$$\|u^{H_1\dots H_n} - u^{\#}\|_{L^p} \leq \|u^{H_1\dots H_n} - \phi^{H_1\dots H_n}\|_{L^p} + \|\phi^{H_1\dots H_n} - \phi^{\#}\|_{L^p} + \|\phi^{\#} - u^{\#}\|_{L^p}$$
$$\leq \|u - \phi\|_{L^p} + \|\phi^{H_1\dots H_n} - \phi^{\#}\|_{L^p} + \|\phi - u\|_{L^p} \leq 3\epsilon.$$

To get to the last line of this computation, we implicitly made a reiterated use of Theorems 3.18 and a direct use of Theorem 3.24 to assert that $\|u^{H_1\dots H_N} - \phi^{H_1\dots H_n}\|_{L^p} \leq \|u - \phi\|_{L^p}$ and $\|u^{\#} - \phi^{\#}\|_{L^p} \leq \|u - \phi\|_{L^p}$. □

If u is in $W^{1,p}(\mathbb{R}^N)$, we can show that $u^{\#} \in W^{1,p}$ and that one can approximate, with respect to the weak topology of L^p, its gradient by gradients of iterated polarizations of u. This is the content of the next theorem, which will be instrumental in the next section (see Theorem 3.34).

Theorem 3.27. Let $u \in W^{1,p}(\mathbb{R}^N, \mathbb{R}^+)$. Then $u^{\#} \in W^{1,p}(\mathbb{R}^N)$, and there exists a sequence $(u_n) \subset P_u$ such that

$$u_n \to u^{\#} \text{ in } L^p, \text{ and } \nabla u_n \rightharpoonup \nabla u^{\#} \text{ in } L^p.$$

Before proceeding to the proof of this theorem, let us observe that it has a direct, yet important consequence.

Corollary 3.28. We have

$$u^{\#} \in W^{1,p}(\mathbb{R}^N) \text{ for all } u \in W^{1,p}(\mathbb{R}^N).$$

To prove the corollary, we simply observe that, for every $u \in W^{1,p}(\mathbb{R}^N)$, $|u| \in W^{1,p}(\mathbb{R}^N)$. Therefore, $u^{\#} = (|u|)^{\#}$ is in $W^{1,p}(\mathbb{R}^N)$.

To prove Theorem 3.27, we rely on the fact that, for any sequence $(u_n) \subset P_u$, $|\nabla u_n| = |\nabla u| \circ \Phi_n$ and $u_n = u \circ \Phi_n$ with Φ_n measure-preserving and bijective (thm 3.16), and on the following

Observation 3.29. Let $u \in W^{1,1}(\mathbb{R}^N)$.
For all $\epsilon > 0$, there exists $\delta > 0$ such that, for any A measurable,

$$\int_A |u|dx > \int_{\mathbb{R}^N} |u|dx - \delta \implies \int_{A^c} |\nabla u|dx < \epsilon.$$

This is implied by the fact that $|\nabla u|dx << |u|dx$ (see [9, Theorem 6.11]). And, indeed, $|\nabla u|dx << |u|dx$ as a direct consequence of the following property of the gradient of a Sobolev function

$$\nabla u = 0 \text{ a.e. in } \{x \in \mathbb{R}^N, u = 0\}.$$

See Theorem 4 in [5, Chapter 4] for a proof of this fact.

Proof of Theorem 3.27. Consider a sequence $(u_n) \subset P_u$ that converges in L^p to u (see Theorem 3.26).

This sequence verifies $u_n \in W^{1,p}$ and $\|\nabla u_n\|_{L^p} = \|\nabla u\|_{L^p}$ for all n.
In the case where $p > 1$, weak compactness in L^p gives the conclusion, as one can extract a weakly converging subsequence of (∇u_n), and identify limits (the identification procedure is written down below).

When $p = 1$, one needs to show that we are dealing with a relatively weakly compact sequence. For this purpose, we rely on the Dunford-Pettis criterion (see [3, p.115]).

In the light of this criterion, proving that (∇u_n) is a relatively weakly compact reduces to the verification of the following two facts.

- (Equi-integrability) For all $\epsilon > 0$, there exists $\delta > 0$ such that for all A measurable, $|A| < \delta$,
$$\int_A |\nabla u_n| \, dx < \epsilon, \text{ for all } n \in \mathbb{N}.$$
- (Concentration) For all $\epsilon > 0$, there exists $A > 0$ such that
$$\int_{\mathbb{R}^N \setminus B(0,A)} |\nabla u_n| \, dx < \epsilon, \text{ for all } n \in \mathbb{N}.$$

Let us proceed to the proof of these facts.

- (Equi-integrability) Let $\epsilon > 0$. There exists $\delta > 0$ such that for all Borel set A
$$|A| < \delta \implies \int_A |\nabla u| \, dx < \epsilon.$$
Therefore, for all Borel sets A such that $|A| < \delta$, one finds that
$$\int_A |\nabla u_n| \, dx = \int_{\Phi_n(A)} |\nabla u| \, dx < \epsilon, \text{ since } |\Phi_n(A)| < \delta.$$
- (Concentration) Let $\epsilon > 0$ and consider $\delta > 0$ given by Observation 3.29 applied to u. There exists N such that, for all $n \geq N$, for all $B \subset \mathbb{R}^N$,
$$\int_B u_n dx > \int_B u \, dx - \frac{\delta}{2}.$$
Therefore, if we consider $A_N > 0$ such that $\int_{B(0,A_N)^c} u \, dx < \frac{\delta}{2}$, we find that for all $n \geq N$
$$\int_{\Phi_n(B(0,A_N))} u \, dx = \int_{B(0,A_N)} u_n dx > \int_{B(0,A_N)} u \, dx - \frac{\delta}{2} > \int_{\mathbb{R}^N} u \, dx - \delta.$$
This gives, by Observation 3.29, that for all $n \geq N$,
$$\int_{B(0,A_N)} |\nabla u_n| \, dx = \int_{\Phi_n(B(0,A_N))} |\nabla u| \, dx < \epsilon.$$
On the other hand, for each $n < N$, one finds $R_n > 0$ such that
$$\int_{\mathbb{R}^N \setminus B(0,R_n)} |\nabla u_n| \, dx < \epsilon.$$
Taking the maximum between those numbers $(R_n)_{n<N}$ and A_N, one has the desired conclusion.

In conclusion, by the Dunford-Pettis criterion, the sequence (∇u_n) is relatively weakly compact in L^1. Therefore, one can extract a subsequence (∇u_{n_k}) that converges weakly in L^1 to a function $G \in L^1$. However, u_n converges to $u^\#$ in L^1, thus (∇u_{n_k}) also converges to $\nabla u^\#$ in the sense of distributions. This implies that $\nabla u^\#$ is a function in L^1.

Therefore, $u^\#$ is in $W^{1,1}(\mathbb{R}^N)$ and (u_{n_k}) verifies the conclusion of the theorem. □

3.5 Spherically symmetric rearrangement in Sobolev spaces and BV

The goal of this section is to use the results of the previous one in order to estimate, in various contexts, the gradient of the rearrangement of a Sobolev function. Throughout this section, we will use the classical:

Proposition 3.30. Let $(f_n) \in BV(\Omega)$ such that $f_n \to f$ in $L^1_{loc}(\Omega)$. Then

$$|Df|(\Omega) \leq \liminf_n |Df_n|(\Omega).$$

Theorem 3.31. (Concentration inequalities)
Let $1 \leq p < \infty$ and let $u \in W^{1,p}(\mathbb{R}^N)$. Consider $g : \mathbb{R}^+ \to \mathbb{R}$ a convex function that verifies, for some $C > 0$,

$$g(s) \leq C(1 + s^p), \text{ for all } s \geq 0.$$

Then

$$\sup_{|A|=t} \left\{ \int_A g(|\nabla u^\#|)\, dx \right\} \leq \sup_{|A|=t} \left\{ \int_A g(|\nabla u|)\, dx \right\}, \text{ for all } t > 0.$$

In the particular case where $g = |\cdot|^p$, one finds that

$$\sup_{|A|=t} \left\{ \int_A |\nabla u^\#|^p\, dx \right\} \leq \sup_{|A|=t} \left\{ \int_A |\nabla u|^p\, dx \right\}, \text{ for all } t > 0.$$

The proof of Theorem 3.32 relies on the following straightforward

Observation 3.32. Let $1 \leq p < \infty$, and let $A \subset \mathbb{R}^N$ be a Borel set of finite Lebesgue measure. We consider a convex function $g : \mathbb{R}^+ \to \mathbb{R}$ such that $|g(t)| \leq C(1 + |t|^p)$ and define a functional $\mathcal{G}$ on $L^p(A)$ by

$$\mathcal{G}(f) := \int_A g(|f|)\, dx, \text{ for all } f \in L^p(A).$$

$\mathcal{G}$ is well-defined and is weakly l.s.c.

Proof of Theorem 3.31. Consider g as in Theorem 3.31 and Observation 3.32. We only need to prove the concentration inequalities for $u \in W^{1,p}$, almost everywhere nonnegative, since this gives immediately the general case.

One approximates $u^\#$ by a sequence (u_n) of iterated polarizations of u, as in Theorem 3.27. Keep in mind that, by Theorem 3.16, $|\nabla u_n| = |\nabla u| \circ \Phi_n$, with Φ_n bijective and measure-preserving.

Consider $t > 0$, and a Borel set $A \subset \mathbb{R}^N$ of measure $|A| = t$.

$$\begin{aligned} \int_A g(|\nabla u_n|)\, dx = \int_A g(|\nabla u| \circ \Phi_n)\, dx &= \int_{\Phi_n(A)} g(|\nabla u|)\, dx \\ &\leq \sup_{|B|=t} \left\{ \int_B g(|\nabla u|)\, dx \right\} \end{aligned}$$

We are in position to apply Observation 3.32, which gives that $\mathcal{G} := \int_A g(|\nabla \cdot|)\, dx$ is l.s.c. with respect to the weak topology of $L^p(A)$. Since $\nabla u_n \rightharpoonup \nabla u^\#$ in L^p, we find that

$$\int_A g(|\nabla u^\#|)\, dx \leq \liminf_n \int_A g(|\nabla u_n|)\, dx \leq \sup_{|B|=t} \left\{ \int_B g(|\nabla u|)\, dx \right\}$$

and the desired result, passing to the sup on all Borel sets A such that $|A| = t$. □

We obtain a similar result in BV, by approximating any $f \in BV$ with a sequence in $(f_n) \subset W^{1,1}$, with $|Df_n|(\mathbb{R}^N) \to |Df|(\mathbb{R}^N)$, and using proposition 3.30.

Corollary 3.33. Let $u \in BV(\mathbb{R}^N)$. Then $u^\# \in BV(\mathbb{R}^N)$ with $\|Du^\#\|(\mathbb{R}^N) \leq \|Du\|(\mathbb{R}^N)$.

We next present a result in the vein of Theorem 3.31 and of Corollary 3.33.

Theorem 3.34. Let $u \in W^{1,p}(\mathbb{R}^N)$ and $1 \leq p < \infty$. Let $F : \mathbb{R}^+ \times \mathbb{R}^+ \to \mathbb{R}^+$ be such that:

- F is continuous
- $\zeta \to F(\omega, \zeta)$ is convex for all $\omega \in \mathbb{R}^+$.

Then:

$$\int_{\mathbb{R}^N} F(u^\#, |\nabla u^\#|)\, dx \leq \int_{\mathbb{R}^N} F(|u|, |\nabla u|)\, dx$$

The proof of Theorem 3.34 is identical to the proof we presented of Theorem 3.31. In order to proceed, one uses the following result, in the same way we used Observation 3.32 in the proof of Theorem 3.31.

Theorem 3.35. ([4, Theorem 3.23]) Let Ω be an open set in $\mathbb{R}^N$, and $1 \leq p, q < \infty$.
Let $F : \Omega \times \mathbb{R} \times \mathbb{R}^N \to \mathbb{R}^+$ be a measurable function such that:

- $(\xi, \zeta) \to F(x, \xi, \zeta)$ continuous for a.e $x \in \Omega$.
- $\zeta \to F(x, \xi, \zeta)$ convex, for a.e $x \in \Omega$, and every $\xi \in \mathbb{R}$.

Then:

$$\left[\xi_n \to \xi \in L^p(\mathbb{R}^N)\,,\ \zeta_n \rightharpoonup \zeta \in L^q(\mathbb{R}^N)\right]$$
$$\implies \int_{\mathbb{R}^N} F(x, \xi(x), \zeta(x))dx \leq \liminf_n \int_{\mathbb{R}^N} F(x, \xi_n(x), \zeta_n(x))dx.$$

Finally, we mention that one can prove, also using Theorem 3.35, the exact generalization of Theorem 3.31 to the case where the function g in Theorem 3.31 depends on two variables, one for the gradient of a Sobolev function, the other for the function itself.

3.6 Rearrangement is continuous in $W^{s,p}$

Theorem 3.36. For any any measurable function $u : \mathbb{R}^N \to \mathbb{R}$, we have

$$\int_{\mathbb{R}^N \times \mathbb{R}^N} \frac{|u^\#(y) - u^\#(x)|^p}{\|y - x\|^{sp+N}} dxdy \leq \int_{\mathbb{R}^N \times \mathbb{R}^N} \frac{|u(y) - u(x)|^p}{\|y - x\|^{sp+N}} dxdy.$$

Moreover, we have

$$u_n \to u \text{ in } W^{s,p}(\mathbb{R}^N) \implies u_n^\# \to u^\# \text{ in } W^{s,p}(\mathbb{R}^N).$$

Proof ot theorem 3.36. The fact that the rearrangement does not increase the seminorm is a direct application of Theorem 3.22 with $k : (0,\infty) \ni t \to \dfrac{1}{t^{sp+N}}$. Let us focus on the continuity of the spherically symmetric rearrangement.

Consider a sequence (u_n) in $W^{s,p}$ such that $u_n \xrightarrow[n\to\infty]{W^{s,p}} u$. We aim at showing that

$$|u_n^\# - u^\#|^p_{W^{s,p}} = \int_{\mathbb{R}^N\times\mathbb{R}^N} \frac{|(u_n^\# - u^\#)(y) - (u_n^\# - u^\#)(x)|^p}{\|y-x\|^{sp+N}}\,dxdy \xrightarrow[n\to\infty]{} 0.$$

For all $\delta > 0$, we have

$$\begin{aligned}|u_n^\# - u^\#|^p_{W^{s,p}} &= \int_{\|x-y\|<\delta} \frac{|(u_n^\# - u^\#)(y) - (u_n^\# - u^\#)(x)|^p}{\|y-x\|^{sp+N}}\,dxdy \\ &\quad + \int_{\|x-y\|\geq\delta} \frac{|(u_n^\# - u^\#)(y) - (u_n^\# - u^\#)(x)|^p}{\|y-x\|^{sp+N}}\,dxdy\end{aligned}$$

$$|u_n^\# - u^\#|^p_{W^{s,p}} := I_{<\delta} + I_{\geq\delta}.$$

One only needs to prove the following.

Claim : (1) For all $\delta > 0$, $I^n_{\geq\delta} \xrightarrow[n\to\infty]{} 0$.

(2) Given $\epsilon > 0$, there exists $\delta > 0$ such that $I^n_{<\delta} \leq \epsilon$, for n sufficiently large.

Proof of (1) : Let $\delta > 0$. For all $n \in \mathbb{N}$,

$$\begin{aligned}I^n_{\geq\delta} &\leq 2^{p-1}\left\{\int_{|x-y|\geq\delta} \frac{|u_n^\#(x) - u^\#(x)|^p}{\|y-x\|^{sp+N}}\,dxdy + \int_{|x-y|\geq\delta} \frac{|u_n^\#(y) - u^\#(y)|^p}{\|y-x\|^{sp+N}}\,dxdy\right\} \\ &\leq 2^p \int_{x\in\mathbb{R}^N} |u^\#(x) - u_n^\#(x)|^p \left\{\int_{\|y-x\|\geq\delta} \frac{1}{\|x-y\|^{sp+N}}dy\right\}dx\end{aligned}$$

by Fubini's theorem. Therefore, computing the integral in the y variable, one finds

$$\begin{aligned}I^n_{\geq\delta} &\leq \frac{2^p\beta_N}{sp\delta^{sp}} \int_{x\in\mathbb{R}^N} |u^\#(x) - u_n^\#(x)|^p dx \\ &\xrightarrow[n\to\infty]{} 0,\end{aligned}$$

since $u_n^\# \to u^\#$ in $L^p(\mathbb{R}^N)$, by Theorem 3.24, since $u_n \to u$ in $L^p(\mathbb{R}^N)$.

Proof of (2) : Let $\epsilon > 0$. An application of dominated convergence implies that there exists δ such that

$$\int_{\|x-y\|<\delta} \frac{|u(y) - u(x)|^p}{\|y-x\|^{sp+N}}\,dxdy < \frac{\epsilon}{2}, \tag{3.4}$$

since $(x,y) \to \dfrac{|u(y) - u(x)|^p}{\|y-x\|^{sp+N}}$ is in $L^1(\mathbb{R}^N \times \mathbb{R}^N)$. On the other hand,

$$\int_{\|x-y\|<\delta} \frac{|u_n(y) - u_n(x)|^p}{\|y-x\|^{sp+N}}\,dxdy \xrightarrow[n\to\infty]{} \int_{\|x-y\|<\delta} \frac{|u(y) - u(x)|^p}{\|y-x\|^{sp+N}}\,dxdy,$$

because $|u_n - u|_{W^{s,p}(\mathbb{R}^N)} \to 0$. Therefore, for n sufficiently large,

$$\int_{\|x-y\|<\delta} \frac{|u_n(y)-u_n(x)|^p}{\|y-x\|^{sp+N}}\,dxdy < \int_{\|x-y\|<\delta} \frac{|u(y)-u(x)|^p}{\|y-x\|^{sp+N}}\,dxdy + \frac{\epsilon}{2} < \epsilon.$$

On the other hand, since $k(x) = \dfrac{\mathbb{1}_{[0,\delta)}(x)}{x^{sp+N}}$ is a non-increasing and nonnegative function defined on $\mathbb{R}^+$, an application of Theorem 3.22 gives that, for any function $f \in L^p(\mathbb{R}^N)$,

$$\int_{\mathbb{R}^N\times\mathbb{R}^N} |f^\#(y)-f^\#(x)|^p \frac{\mathbb{1}_{[0,\delta)}(\|x-y\|)}{\|x-y\|^{sp+N}}\,dxdy \le \int_{\mathbb{R}^N\times\mathbb{R}^N} |f(y)-f(x)|^p \frac{\mathbb{1}_{[0,\delta)}(\|x-y\|)}{\|x-y\|^{sp+N}}\,dxdy.$$

We apply this observation to the right hand side of the following estimate, i.e., to all the terms of the sequence (u_n) and to u. For all $n \in \mathbb{N}$,

$$I_{<\delta} \le 2^{p-1} \int_{\|x-y\|<\delta} \frac{|u_n^\#(y)-u_n^\#(x)|^p}{\|y-x\|^{sp+N}} \mathbb{1}_{B(0,\delta)}(x-y)dxdy + 2^{p-1} \int_{\|x-y\|<\delta} \frac{|u^\#(y)-u^\#(x)|^p}{\|y-x\|^{sp+N}} \mathbb{1}_{B(0,\delta)}(x-y)dxdy$$

and, therefore,

$$I_{<\delta} \le 2^{p-1} \int_{\|x-y\|<\delta} \frac{|u_n(y)-u_n(x)|^p}{\|y-x\|^{sp+N}} \mathbb{1}_{B(0,\delta)}(x-y)dxdy + 2^{p-1} \int_{\|x-y\|<\delta} \frac{|u(y)-u(x)|^p}{\|y-x\|^{sp+N}} \mathbb{1}_{B(0,\delta)}(x-y)dxdy$$

With $\delta > 0$ chosen such that (3.4) holds, one gets that

$$I^n_{<\delta} \le 2^p\epsilon, \text{ for all } n \text{ sufficiently large.}$$

This implies the second part of the claim and completes the proof of Theorem 3.36. □

4 Real interpolation

4.1 The example of the Marcinkiewicz theorem

As mentioned in the Overview, Marcinkiewicz' proof of Theorem 2.2 was one of the main inspirations in the elaboration of the real interpolation theory, since it shows how to construct a concrete interpolation space.

Given two normed spaces X_0 and X_1, we say that they form a compatible couple (X_0, X_1) if they are both embedded in a same Hausdorff topological vector space. In this case, one can consider the spaces

$$X_0 + X_1 \text{ equipped with the norm } \|x\|_{X_0+X_1} := \inf\{\|x_0\|_{X_0} + \|x_1\|_{X_1}\,,\ x = x_0 + x_1\}$$

$X_0 \cap X_1$ equipped with the norm $\|x\|_{X_0 \cap X_1} := \max\{\|x\|_{X_0}, \|x_1\|_{X_1}\}$.

In this setting, we say that a normed space X is an interpolation space if

$$X_0 \cap X_1 \hookrightarrow X \hookrightarrow X_0 + X_1$$

and if

$$T \in B(X_0, X_0) \cap B(X_1, X_1) \implies T \in B(X, X),$$

for all linear operator T defined on $X_0 + X_1$. This definition corresponds to the type of spaces that we consider in "interpolation" results. They are lying between two endpoint spaces and give automatic bounds on operators that are bounded on these endpoint spaces (here X_0 and X_1).

For example, the Marcinkiewicz theorem (2.2) shows that L^p is an interpolation space, relatively to the endpoint spaces L^{p_0} and L^{p_1}.

It is easy to see that (L^{p_0}, L^{p_1}) is a compatible couple, and that

$$L^{p_0} \cap L^{p_1} \hookrightarrow L^p \hookrightarrow L^{p_0} + L^{p_1}.$$

The Marcinkiewicz theorem assures automatic bounds on operators that are in $B(L^{p_0}, L^{p_0})$ and $B(L^{p_1}, L^{p_1})$. One of the main features of this theorem is that it asks, in fact, for much weaker assumptions, as one only needs an operator to be in $B(L^{p_0}, L^{p_0,\infty}) \cap B(L^{p_1}, L^{p_1,\infty})$ to assure $L^p - L^p$ boundedness.

But what is really interesting with respect to the study of interpolation is that Theorem 2.2 gives us a concrete example of an interpolation space.

Given $f \in L^p$, the proof of Theorem 2.2 relies on varying decompositions of $f = f_{0,t} + f_{1,t}$ in $L^{p_0} + L^{p_1}$.

$f_{0,t} \in L^{p_0}$ and $f_{1,t} \in L^{p_1}$ are chosen in order to be able to establish estimates of the form :

$$\int_0^\infty F_0(t, \|f_{0,t}\|_{L^{p_0}})\, dt \leq C_0 \|f\|_{L^p}^p \ , \ \int_0^\infty F_1(t, \|f_{1,t}\|_{L^{p_1}})\, dt \leq C_1 \|f\|_{L^p}^p.$$

The proof one finds in [10] is based on decomposing f as $f = f\mathbb{1}_{|f|<a} + f\mathbb{1}_{|f|\geq a}$. A finer estimate consists in taking $a = f^*(t)$ to proceed. It allows us to decompose $f = f_{0,t} + f_{1,t}$, with $f_{0,t}$ and $f_{1,t}$ satisfying

$$\int_0^\infty \frac{1}{t^{\sigma p+1}} \left\{\|f_{0,t}\|_{L^{p_0}} + t\|f_{1,t}\|_{L^{p_1}}\right\}^p dt \leq C\|f\|_{L^p}^p \tag{4.1}$$

(see [2, p.10]).

The $K-$method of interpolation is an abstraction of this feature of L^p spaces. It leads to the construction of spaces that are tailor-made to verify Theorem 2.5.

4.2 The interpolation space $(X_0, X_1)_{\sigma,q}$

4.2.1 The K-Method

Throughout Section 4.2, we will consider a compatible couple (X_0, X_1) of normed spaces.

Let us consider, for each $t > 0$ the norm $K(t, \cdot, X_0, X_1)$ on $X_0 + X_1$ defined by

$$K(t, x, X_0, X_1) := \inf\{\|x_0\|_{X_0} + t\|x_1\|_{X_1}, \text{ with } x_0 \in X_0,\ x_1 \in X_1,\ x = x_0 + x_1\}.$$

The following properties are straightforward. 1. $K(t,\cdot,X_0,X_1)$ is equivalent to $\|\cdot\|_{X_0+X_1}$. 2. $K(\cdot,x,X_0,X_1)$ is non-decreasing, concave and such that

$$K(t,x,X_0,X_1) \leq \max\{1,t/\tau\}K(\tau,x,X_0,X_1)$$

for all $t,\tau \in \mathbb{R}^+ \setminus \{0\}, x \in X_0+X_1$.
See [7], pp.518-520, for a survey of the most useful properties of K.

We will omit the reference to the compatible couple that we consider when it is obvious and use the notation $K(t,x) := K(t,x,X_0,X_1)$.

Equipped with this so-called "K-functional", we consider the following spaces.

Definition 4.1. Let $0<\sigma<1$ and $1 \leq q \leq \infty$.
We define $(X_0,X_1)_{\sigma,q}$ to be the vector space of all $x \in X_0+X_1$ such that

$$\int_0^\infty \left(\frac{K(t,x)}{t^\sigma}\right)^q \frac{dt}{t} < \infty,$$

equipped with the norm $\|x\|_{\sigma,q} := \left\{\int_0^\infty \left(\frac{K(t,x)}{t^\sigma}\right)^q \frac{dt}{t}\right\}^{1/q}$ when $q<\infty$, and with the modification $\|x\|_{\sigma,\infty} := \sup_{t>0} \frac{K(t,x)}{t^\sigma} < \infty$ when $q=\infty$.

Remark 4.2. In the above definition, one may consider, more generally, semi-normed spaces. The results we present in subsection 4.2.1 have straightforward semi-normed counterparts.

The space defined above is an interpolation space, as it verifies

$$X_0 \cap X_1 \hookrightarrow (X_0,X_1)_{\sigma,q} \hookrightarrow X_0+X_1$$

and Theorem 2.5 (see [6], Theorems 16.11 and 16.12). Morevover, if X_0 and X_1 are Banach spaces, then $(X_0,X_1)_{\sigma,q}$ is also a Banach space.

Let us mention an important condition for a vector to be in $(X_0,X_1)_{\sigma,q}$.

Proposition 4.3. Let $0<\sigma<1$ and $1 \leq q < \infty$. Then

$$K(t,x,X_0,X_1) \leq (\sigma q)^{1/q} t^\sigma \|x\|_{\sigma,q}, \text{ for all } x \in (X_0,X_1)_{\sigma,q}.$$

Before studying the features of $(X_0,X_1)_{\sigma,q}$ and their applications, we must investigate another way to realise this space.

4.2.2 The equivalent J-method

Let $0<\sigma<1$ and $1 \leq q < \infty$ and consider $x \in (X_0,X_1)_{\sigma,q}$.

Proposition 4.3 implies that x can be decomposed in X_0+X_1 according to a certain "balance" in the repartition of mass between the X_0 and X_1 norms. Indeed, it implies, combined with the definition of K, that for every $t>0$, there exists $(x_t,y_t) \in X_0 \times X_1$ such that $x = x_t + y_t$ and

$$\|x_t\|_{X_0} + t\|y_t\|_{X_1} \leq 2(\sigma q)^{1/q} t^\sigma \|x\|_{\sigma,q}. \tag{4.2}$$

Thus, these decompositions $x = x_t + y_t$ verify

$$\|x-y_t\|_{X_0+X_1} = \|x_t\|_{X_0+X_1} \leq \|x_t\|_{X_0} = \mathcal{O}(t^\sigma) \text{ as } t \to 0^+;$$

$$\|x - x_t\|_{X_0+X_1} = \|y_t\|_{X_0+X_1} \leq \|y_t\|_{X_1} = \mathcal{O}(\frac{1}{t^{1-\sigma}}) \text{ as } t \to \infty.$$

Therefore, considering the sequence $(x_{2^k})_{k\in\mathbb{Z}}$, one finds that

$$\sum_{k=-l}^{k=l} (x_{2^k} - x_{2^{k-1}}) = x_{2^l} - x_{2^{-l-1}} \xrightarrow[l\to\infty]{} x \text{ in } X_0 + X_1$$

On the other hand, for all $k \in \mathbb{Z}$,

$$x_{2^k} - x_{2^{k-1}} = y_{2^{k-1}} - y_{2^k} \in X_0 \cap X_1,$$

since $x = x_t + y_t$ for all $t > 0$. Therefore, we have shown that $x \in (X_0, X_1)_{\sigma,q}$ is a sum converging in $X_0 + X_1$ of vectors in $X_0 \cap X_1$ (in a sense that is made precise in this section).

The J-method of interpolation, dual to the K-construction, is based on this phenomenon. For every $x \in X_0 \cap X_1$, we define $J(\cdot, x, X_0, X_1)$ by

$$J(t, x, X_0, X_1) := \max\{\|x\|_{X_0}, t\|x\|_{X_1}\}, \text{ for all } t > 0.$$

For every $t > 0$, $J(t, \cdot)$ is a norm on $X_0 \cap X_1$, equivalent to $\|\cdot\|_{X_0\cap X_1}$. On the other hand, $J(\cdot, x)$ is a non decreasing concave function, for every $x \in X_0 \cap X_1$. See [6], p.526 and [2], Lemma 3.2.1 and Theorem 3.2.2, to learn about some of the most important properties of J.

We now consider a functional on X_0+X_1 that will allow us to define an interpolation space. Considering $x \in X_0 + X_1$, we will say that a measurable function $u : (0, \infty) \to X_0 \cap X_1$ is (J, x)-admissible (sometimes we will say that u is a representative of x) if :

- $(0, \infty) \ni t \to \frac{u(t)}{t}$ is in $L^1((0, \infty); X_0 + X_1)$ and $L^1_{loc}((0, \infty); X_0 \cap X_1))$.
- $x = \displaystyle\int_0^\infty \frac{u(t)}{t} dt$, in the sense of the Bochner integral w.r.t. $\|\cdot\|_{X_0+X_1}$.

Given this notion of representation, we define, for $0 < \sigma < 1$ and $1 \leq q \leq \infty$,

$$\|x\|_{\sigma,q,J} := \begin{cases} \inf\left\{ \left\{ \int_0^\infty \left(\frac{J(t,u(t))}{t^\sigma} \right)^q \frac{dt}{t} \right\}^{1/q}, \ u \text{ is } (J,x) - \text{admissible} \right\} & \text{if } q < \infty \\ \inf\left\{ \sup_{t>0} \left\{ \frac{J(t,u(t))}{t^\sigma} \right\}, \ u \text{ is } (J,x) - \text{admissible} \right\} & \text{if } q = \infty \end{cases}$$

and the space $(X_0, X_1)_{\sigma,q,J}$ by

$$(X_0, X_1)_{\sigma,q,J} := \{x \in X_0 + X_1 \, ; \, \|x\|_{\sigma,q,J} < \infty\}.$$

$(X_0, X_1)_{\sigma,q,J}$ is constructed so as to be characterized by a property similar to the property of $(X_0, X_1)_{\sigma,q}$ we described at the beginning of this section. This is content of the following.

Proposition 4.4. Let $0 < \sigma < 1$ and $1 \leq q \leq \infty$. $x \in X_0 + X_1$ belongs to $(X_0, X_1)_{\sigma,q}$ if and only if there exists a sequence $(x_k)_{k\in\mathbb{Z}}$ such that

$$(1) \sum_{k\in\mathbb{Z}} \|x_k\|_{X_0+X_1} < \infty \ ; \ (2) \ x = \sum_{k\in\mathbb{Z}} x_k \ ; \ (3) \ \sum_{k\in\mathbb{Z}} \left\{ 2^{-k\sigma} J(2^k, x_k) \right\}^q < \infty.$$

Moreover the set $\mathcal{R}$ of all sequences (x_k) that verify the properties $(1)-(3)$ allow us to consider a norm given by

$$x \to \inf_{(x_k)\in\mathcal{R}} \{ \left\| \{2^{-k\sigma} J(2^k, a_k)\right\|_{l^q(\mathbb{Z})} \}.$$

This norm is equivalent to $\|\cdot\|_{\sigma,q,J}$. This equivalence of norms implies, in particular, that when $q < \infty$, a sequence in $\mathcal{R}$ verifies

$$\left\| x - \sum_{|k|\leq M} x_k \right\|_{\sigma,q,J} \longrightarrow 0 \text{ as } M \to \infty.$$

Thus, we find that

$$X_0 \cap X_1 \text{ is dense in } (X_0, X_1)_{\sigma,q,J} \text{ if } q < \infty. \tag{4.3}$$

The introduction to subsection 4.2.2 illustrates how the property 4.3 could also be true with $(X_0, X_1)_{\sigma,q,J}$ replaced by $(X_0, X_1)_{\sigma,q}$. We have no idea of a direct proof of this fact, but it turns out there is no need for one. Indeed, the spaces constructed with the two methods are the same.

Theorem 4.5. ([6, p.527])
Let $0 < \sigma < 1$ and $1 \leq q \leq \infty$. There exists a constant $c > 0$ such that

$$c^{-1}\|x\|_{\sigma,q,J} \leq \|x\|_{\sigma,q} \leq c\|x\|_{\sigma,q,J}, \text{ for every } x \in X_0 + X_1.$$

Therefore, $(X_0, X_1)_{\sigma,q} = (X_0, X_1)_{\sigma,q,J}$, with equivalence of norms.

4.2.3 Lorentz spaces

In subsection 4.2.3, all the statements are to be understood with respect to a measure space (R, μ).

By Proposition 3.4, we have that $\|f\|_{L^p}^p = \displaystyle\int_0^\infty \left(f^*(t)t^{1/p}\right)^q \frac{dt}{t}$ for all $f \in M(R, \mu)$, when $q = p$. Allowing the "microscopic" parameter q to vary in $(0, \infty)$, we find a new range of function spaces. Using these spaces, instead of Lebesgue spaces only, one may be able to obtain sharper estimates of the "sizes" of functions.

Definition 4.6. (Lorentz spaces) Let $0 < p < \infty$ and $0 < q \leq \infty$.
We consider the function space defined by

$$L^{p,q}(R, \mu) := \{f \in M_0(R, \mu) \text{ such that } ||f||_{L^{p,q}(R,\mu)}^q := \int_0^\infty (f^*(t)t^{1/p})^q \frac{dt}{t} < \infty\}$$

when q is finite, and with the modification $||f||_{L^{p,\infty}} := \sup_{t>0}\left\{ f^*(t)t^{1/p} \right\}$, when $q = \infty$.

Remark 4.7. Another classical way to define Lorentz spaces relies on the distribution function instead of the decreasing rearrangement. More specifically, we consider the quasi-norms

$$|f|_{L^{p,q}} := \left\{ p \int_0^\infty (\mu_f(t)^{1/p}\, t)^q \frac{dt}{t} \right\}^{1/q} \text{ when } q < \infty$$

and

$$|f|_{L^{p,\infty}} := \sup_{t>0} \mu_f(t)^{1/p} t.$$

These functionals, defined on the set of measurable functions $M(R,\mu)$, are in fact equal to their counterpart defined via the decreasing rearrangement. We prove this fact in the case where $q < \infty$; the remaining case is similar.

Let us consider a simple function $f : (R,\mu) \to [0,\infty]$. There exists $m \in \mathbb{N}^*$, a disjoint family of $\mu-$measurable sets $(A_i)_{1\leq i\leq m}$ and numbers $a_1 > a_2 > \cdots > a_m > 0$ such that

$$f = \sum_{i=1}^{m} a_i \mathbb{1}_{A_i}.$$

One easily calculates the distribution function and the decreasing rearrangement of such a function. Let us denote $S_i := \sum_{j=1}^{i} \mu(A_j)$ for all $1 \leq i \leq m$, $S_0 := 0$, and $a_{m+1} := 0$. We have

$$\mu_f(t) = \sum_{i=1}^{m} S_i \mathbb{1}_{[a_{i+1},a_i)} \text{ and } f^* = \sum_{i=1}^{m} a_i \mathbb{1}_{[S_{i-1},S_i)}$$

Therefore, we have that

$$\int_0^\infty \left(f^*(t)t^{1/p}\right)\, dt = \int_0^\infty \left\{ \left(\sum_{i=1}^{m} a_i \mathbb{1}_{[S_{i-1},S_i[}(t) \right) t^{1/p} \right\}^q \frac{dt}{t} = \sum_{i=1}^{m} a_i^q \int_{S_{i-1}}^{S_i} t^{q/p} \frac{dt}{t}$$

and

$$p\int_0^\infty \left(\mu_f(t)^{1/p}t\right)^q \frac{dt}{t} = p\int_0^\infty \left\{ \sum_{i=1}^{m} S_i \mathbb{1}_{[a_{i+1},a_i[} \right\}^{q/p} t^{q-1}\, dt = p\sum_{i=1}^{m} S_i^{q/p} \int_{a_{i+1}}^{a_i} t^{q-1}\, dt.$$

If one of the sets A_i is of infinite measure, or if one of the numbers a_i is infinite, we have that both $\int_0^\infty \left(f^*(t)t^{1/p}\right) \frac{dt}{t}$ and $p\int_0^\infty \left(\mu_f(t)^{1/p}t\right)^q \frac{dt}{t}$ are infinite. On the other hand, if $\mu(A_i) < \infty$ and $0 < a_i < \infty$ for all $1 \leq i \leq m$, we find that

$$\begin{aligned}\int_0^\infty \left(f^*(t)t^{1/p}\right)\, dt &= \frac{p}{q}\sum_{i=1}^{m} a_i^q \left(S_n^{q/p} - S_{n-1}^{q/p}\right) \\ &= \frac{p}{q}\sum_{i=1}^{m} S_i^{q/p}\left(a_i^q - a_{i+1}^q\right) \quad \text{(Abel transform).}\end{aligned}$$

and

$$\begin{aligned}p\int_0^\infty \left(\mu_f(t)^{1/p}t\right)^q \frac{dt}{t} &= p\sum_{i=1}^{m} S_i^{q/p} \int_{a_{i+1}}^{a_i} t^{q-1}\, dt \\ &= \frac{p}{q}\sum_{i=1}^{m} S_i^{q/p}\left(a_i^q - a_{i+1}^q\right)\end{aligned}$$

This computation shows that, for any simple function $f : (R,\mu) \to [0,\infty]$, one has

$$p\int_0^\infty \left(\mu_f(t)^{1/p}t\right)^q \frac{dt}{t} = \int_0^\infty \left(f^*(t)t^{1/p}\right)\, dt. \tag{4.4}$$

One generalizes identity (4.4) to all measurable functions on (R, μ), by an application of monotone convergence. Indeed, given $f \in M(R, \mu)$, one finds a sequence (f_n) of simple nonnegative functions such that (f_n) is increasing and converges pointwise to $|f|$. In this case, the sequences (μ_{f_n}) and (f_n^*) both are increasing and converging pointwise, to μ_f and f^* respectively (see [1, Propositions 1.3, 1.6]). Therefore, one apply (4.4) to the sequence (f_n) and pass to the limit by monotone convergence, to find that (4.4) holds for f.

We have that $\|\cdot\|_{L^{p,q}(R,\mu)}$ is only a quasi-norm. There exists $c > 0$ such that

$$\|f + g\|_{L^{p,q}} \leq c\{\|f\|_{L^{p,q}} + \|g\|_{L^{p,q}}\}, \text{ for all } f, g \in L^{p,q}(R, \mu). \tag{4.5}$$

However, any quasi-norm that verifies a triangular inequality like (4.5), for a number $c \geq 1$, is equivalent to a norm.

In the case of Lorentz spaces, we show that one of these equivalent norms is the norm of an interpolation space for the couple (L^1, L^∞). Let us begin by calculating the K-functional associated to this couple.

Theorem 4.8. ([1, p.74]) For all $f \in M_0(R, \mu)$, we have that

$$\int_0^t f^*(s)ds = \inf\{||g||_{L^1} + t||h||_{L^\infty},\ g \in L^1,\ h \in L^\infty,\ \text{s.t. } f = g + h\} \text{ for all } t > 0,$$

the two quantities being infinite if f is not in $L^1 + L^\infty$.

A consequence of Theorem 4.8 is the following theorem.

Theorem 4.9. ([6, p.535]) Let $1 < p < \infty$ and $1 \leq q \leq \infty$.
We have that

$$(L^1(R, \mu), L^\infty(R, \mu))_{1-1/p,q} = L^{p,q}(R, \mu)$$

with the interpolation norm being equivalent to $\|\cdot\|_{L^{p,q}(R,\mu)}$.

4.3 Three results, in view of applications

In this section, we present three striking applications of the abstract constructions of the previous section.

From what we have seen of interpolation theory up to this point, it does not seem obvious that it could be useful to simplify the study of concrete spaces.

Let us consider, for example, Lorentz spaces. Theorem 4.9 states that $L^{p,q} = (L^1, L^\infty)_{1-1/p,q}$. However, one may be more interested in realizing all $L^{p,q}$ as interpolation spaces between any L^{p_0,q_0} and L^{p_1,q_1}, as long as $p_0 < p < p_1$, as in Theorem 2.4. For one thing, it would allow us to interpret the Marcinkiewicz theorem as a simple consequence of Theorem 2.5, as one reads in the Overview.

Proceeding methodically, one should show that this result holds for all couples $(L^{p_0,q_0}, L^{p_1,q_1})$, and in order to do so estimate the associated K-functional. The following theorem allows us to avoid such complications.

Theorem 4.10. (Reiteration theorem) Let (X_0, X_1) be an admissible couple of normed spaces. Let $1 \leq q_0, q_1, q \leq \infty$, $0 < \sigma_0 < \sigma_1 < 1$, and $0 < \sigma < 1$. Consider $\theta := (1 - \sigma)\sigma_0 + \sigma\sigma_1$.
1. Set $X_{\sigma_0} := (X_0, X_1)_{\sigma_0,q_0}$, $X_{\sigma_1} := (X_0, X_1)_{\sigma_1,q_1}$. Then

$$(X_0, X_1)_{\theta,q} = (X_{\sigma_0}, X_{\sigma_1})_{\sigma,q}$$

with equivalence of norms.
2. Moreover, we have

$$(X_0, X_{\sigma_0})_{\sigma,q} = (X_0, X_1)_{\sigma\sigma_0,\, q} \text{ and } (X_{\sigma_0}, X_1)_{\sigma,q} = (X_0, X_1)_{(1-\sigma)\sigma_0+\sigma,\, q}$$

with equivalence of norms.

One may find a proof of Theorem 4.10 in [6, Theorem 16.15]. For a proof that only applies to Banach spaces, but that is more transparent, see [2, Theorem 3.5.3].

Theorem 4.10, combined with Theorem 4.9 shows that Theorem 2.4 holds, in the special case where $p_0 > 1$ or $p_0 = q_0 = 1$ (see [2, p.113] for a complete proof of Theorem 2.4).

Real interpolation also allows to systematize and simplify the calculation of the duals of spaces that are interpolation spaces. In order to do so, one proceeds by identification of various dual spaces (see Theorem 4.11) that are incomparable, a priori. One may find the details of this identification in [6, pp.530-535] and the proof of the following theorem.

Theorem 4.11. Let (X_0, X_1) be a compatible couple of Banach spaces, such that $X_0 \cap X_1$ is dense in both X_0 and X_1. Then $(X_0^*, \|\cdot\|_{X_0^*})$ and $(X_1, \|\cdot\|_{X_1^*})$ form a compatible couple, and one can identify $((X_0, X_1)_{\sigma,q})^*$ with $(X_0^*, X_1^*)_{\sigma,q'}$, for all $0 < \sigma < 1$ and $1 \leq q < \infty$ (q' is the Hölder conjugate of q).

Let us finish this section with an observation that leads to a more systematic understanding of extrapolation results like Theorem 2.7.

Theorem 4.12. ([6, p.525]) Let $1 \leq q < \infty$ and $g : [0, \infty) \to [0, \infty)$ be a measurable function. Suppose that there exists $0 < \sigma_0 < 1$ such that

$$\int_0^\infty \frac{[g(t)]^q}{t^{\sigma_0 q+1}}\, dt < \infty.$$

Then one has that

$$\begin{aligned}\liminf_{t\to\infty} g(t) &\leq \liminf_{\sigma\to 0^+} \left\{(\sigma(1-\sigma)q)\int_0^\infty \frac{[g(t)]^q}{t^{\sigma q+1}}\right\}^{1/q} \\ &\leq \limsup_{\sigma\to 0^+} \left\{(\sigma(1-\sigma)q)\int_0^\infty \frac{[g(t)]^q}{t^{\sigma q+1}}\right\}^{1/q} \leq \limsup_{t\to\infty} g(t)\end{aligned}$$

and

$$\begin{aligned}\liminf_{t\to 0^+} \frac{g(t)}{t} &\leq \liminf_{\sigma\to 0^+} \left\{(\sigma(1-\sigma)q)\int_0^\infty \frac{[g(t)]^q}{t^{\sigma q+1}}\right\}^{1/q} \\ &\leq \limsup_{\sigma\to 0^+} \left\{(\sigma(1-\sigma)q)\int_0^\infty \frac{[g(t)]^q}{t^{\sigma q+1}}\right\}^{1/q} \leq \limsup_{t\to\infty} \frac{g(t)}{t}.\end{aligned}$$

Let $x \in (X_0, X_1)_{\sigma_0,q}$ for some $0 < \sigma_0 < 1$. Applying Theorem 4.12 with $g(t) := K(t, x)$, one obtains the following result.

$$\begin{aligned}\lim_{t\to\infty} K(t, x, X_0, X_1) &= \lim_{\sigma\to 0^+} \left[\{\sigma(1-\sigma)\}^{1/q} \|x\|_{\sigma,q}\right], \\ \lim_{t\to 0^+} \frac{K(t, x, X_0, X_1)}{t} &= \lim_{\sigma\to 1^-} \left[\{\sigma(1-\sigma)\}^{1/q} \|x\|_{\sigma,q}\right].\end{aligned}$$

5 Applications to the study of fractional Sobolev Spaces

5.1 Generalities on Besov spaces

Besov spaces generalize fractional Sobolev spaces $W^{s,p}$ with fractional regularity. This generalization allows greater technical flexibility (through the use of the microscopic parameter q below) that can prove useful in several contexts. We will see how it can be used in relation with interpolation theory.

In this section, we survey the most important properties of Besov spaces in view of proving Theorem 5.6, see subsection 5.2.

Definition 5.1. (Difference operators)
Let $h \in \mathbb{R}^N$. We define the difference operator Δ_h on M_0 by

$$\Delta_h f(x) := f(x+h) - f(x), \text{for a.e. } x \in \mathbb{R}^N, \text{ for all } f \in M_0.$$

Higher order difference operators are defined by induction with $\Delta_h^n = \Delta_h \circ \Delta_h^{n-1}$, for all $n \geq 2$. We have the useful identity

$$\Delta_h^n = (\tau_h - Id)^n = \sum_{k=0}^{n} \binom{n}{k} (-1)^{n-k} \tau_{kh}. \tag{5.1}$$

Definition 5.2. (Besov spaces)
Let $1 \leq p, q \leq \infty$, $s > 0$. The inhomogeneous Besov space $B_q^{s,p}$ consists of all functions $f \in L^p(\mathbb{R}^N)$ such that

$$|f|_{B_q^{s,p}}^q := \int_{\mathbb{R}^N} \frac{\|\Delta_h^{\lfloor s \rfloor + 1} f\|_{L^p}^q}{\|h\|^{sq+1}} \, dh < \infty, \text{ if } q < \infty,$$

$$|f|_{B_\infty^{s,p}} = \sup_{h \in \mathbb{R}^N} \frac{\|\Delta_h^{\lfloor s \rfloor + 1} f\|_{L^p}}{\|h\|^s}.$$

The homogeneous Besov space $\dot{B}_q^{s,p}$ consists of all $f \in L^p_{loc}(\mathbb{R}^N)$ such that $|f|_{B_q^{s,p}}$ is finite.

We chose to define Besov spaces with the seminorms $|\cdot|_{B_q^{s,p}}$ in order to point out that they are indeed generalizations of fractional Sobolev spaces. Indeed, one has that $|f|_{B_p^{s,p}} = |f|_{W^{s,p}}$ for any $f \in M_0$, and, therefore, that $B_p^{s,p}(\mathbb{R}^N) = W^{s,p}(\mathbb{R}^N)$ for all $1 \leq p \leq \infty$, and $0 < s < 1$. We now proceed to a survey of the most important (semi)norms that are equivalent to $|\cdot|_{B_q^{s,p}}$, towards our goal of understanding why Besov spaces are interpolation spaces.

The first equivalence (Theorem 5.3) asserts that to compute the Besov norms, one can compute the L^p norms of arbitrarily high order differences. The second (Theorem 5.5) asserts that one only needs to focus on the one dimensional integral of a monotone function, to study Besov norms. This equivalence is particularly useful in relation with Theorem 4.12.

Theorem 5.3. ([6, p.546]) Let $1 \leq p, q \leq \infty$, and $s > 0$.
For every $n \geq \lfloor s \rfloor + 1$, there exists c_1, c_2 only depending on n and s such that, for every measurable $f : \mathbb{R}^N \to \mathbb{R}$ with $|f|_{B_q^{s,p}} < \infty$,

$$c_1 \int_{\mathbb{R}^N} \frac{\|\Delta_h^n f\|_{L^p}^q}{\|h\|^{sp+N}} \, dh \leq |f|_{B_q^{s,p}}^q \leq c_2 \int_{\mathbb{R}^N} \frac{\|\Delta_h^n f\|_{L^p}^q}{\|h\|^{sp+N}} \, dh, \text{ if } q < \infty.$$

When $q = \infty$, we have that

$$c_1 \sup_{t>0} \frac{\|\Delta_h^n f\|_{L^p}}{\|h\|^s} \leq |f|_{B_\infty^{s,p}} \leq c_2 \sup_{t>0} \frac{\|\Delta_h^n f\|_{L^p}}{\|h\|^s}.$$

We now introduce the L^p-moduli of continuity of functions in M_0, which are among the most important tools to study interpolation and extrapolation in Besov spaces.

Definition 5.4. Let $1 \leq p \leq \infty$, $m \in \mathbb{N}^*$.
The m-th order L^p modulus of continuity ω_p^m of a function $f \in M_0$ is defined by

$$\omega_p^m(t,f) := \sup_{\|h\| \leq t} \|\Delta_h^m f\|_{L^p}, \text{ for all } t > 0.$$

A first trivial, but quite useful, observation is that $\omega_p^m(\cdot, f)$ is increasing.

Theorem 5.5. Let $1 \leq p \leq \infty$ and $1 \leq q < \infty$. Let $s > 0$, and $m \in \mathbb{N}$. There exists $c > 0$ depending on N, p, and m such that for every $f \in M_0$,

$$\int_0^\infty [\omega_p^m(t,f)]^q \frac{dt}{t^{sq+1}} \leq \frac{c}{s^p} \int_{\mathbb{R}^N} \frac{\|\Delta_h^m f\|_{L^p}^q}{\|h\|^{N+sq}}\, dh.$$

On the other hand, we have that

$$\int_{\mathbb{R}^N} \frac{\|\Delta_h^m f\|_{L^p}^q}{\|h\|^{N+sq}}\, dh \leq \beta_N \int_0^\infty [\omega_p^m(t,f)]^q \frac{dt}{t^{sq+1}}.$$

When $q = \infty$, one gets similar results, with the natural modification consisting in considering $\sup$ instead of integrals (see [6, pp.548-550]).

We are now in position to understand why Besov spaces are interpolation spaces.

5.2 Besov spaces are interpolation spaces

The setting is the same as in the previous subsection. Let us denote, for $m \in \mathbb{N}^*$,

$$K(t,f) := K(t,f,L^p,W^{m,p}), \text{ for all } f \in L^1_{loc},$$
$$\dot{K}(t,f) := K(t,f,L^p,\dot{W}^{m,p}), \text{ for all } f \in L^1_{loc},$$

with $K(t,f)$ being infinite if $f \notin L^p + W^{m,p}$, respectively $\dot{K}(t,f)$ being infinite if $f \notin L^p + \dot{W}^{m,p}$.

In this subsection, we will study and modify certain aspects of the proof in [1, pp.339-341] of the following theorem.

Theorem 5.6. Let $m \in \mathbb{N}^*$, $1 \leq p \leq \infty$.
1. We have, with finite constants $0 < c_1, c_2 < \infty$ possibly depending on N, m, and p,

$$c_1 K(t^m, f) \leq \omega_p^m(t,f) + t^m \|f\|_{L^p} \leq c_2 K(t^m, f), \text{ for all } 0 < t < 1 \text{ and } f \in L^1_{loc}, \quad (5.2)$$

where all the quantities may be infinite.

As a direct consequence of this estimate, we find that Besov spaces are interpolation spaces:

$$B_q^{\sigma m,p}(\mathbb{R}^N) = (L^p(\mathbb{R}^N), W^{m,p}(\mathbb{R}^N))_{\sigma,q}, \text{ for all } 0 < \sigma < 1,\ 1 \leq q \leq \infty \quad (5.3)$$

with equivalence between the Besov and interpolation norms.
2. Similarly, for homogeneous spaces we have

$$c_1 \dot{K}(t^m, f) \leq \omega_p^m(t,f) \leq c_2 \dot{K}(t^m, f), \text{ for all } f \in L^1_{loc}, \quad (5.4)$$

where all the quantities may be infinite. Therefore,

$$\dot{B}_q^{\sigma m,p}(\mathbb{R}^N) = (L^p(\mathbb{R}^N), \dot{W}^{m,p}(\mathbb{R}^N))_{\sigma,q}, \text{ for all } 0 < \sigma < 1,\ 1 \leq q \leq \infty. \quad (5.5)$$

The most intricate statements in Theorem 5.6 are the first inequality in (5.2), respectively the first inequality in (5.4). The second inequalities in (5.2) and (5.4) are straightforward. So is the fact that (5.2), respectively (5.4), implies (5.3), respectively (5.5); see the detailed proof of Theorem 5.6 in [1, pp. 339-341], and also the discussion below. Therefore, in this subsection we focus on the proof of the first inequality in (5.2), respectively (5.4). More precisely, the two arguments being very similar, we only present a (new) proof of the first inequality in (5.4).

Let us start with the heuristics. In order to estimate the K-functional of a function f in terms of ω_p^m, one has to find a decomposition $f = g_t + h_t$, where the expression of g_t involves $\Delta_h^m f$, respectively the expression of $D^m g_t$ involves $\Delta_h^m f$. An educated guess is $g_t := f - f * \rho_t$, $h_t := f * \rho_t$, where ρ is a mollifier to be specified. When $m = 1$, this approach works for any mollifier ρ. Indeed, in this case, one has

$$f(x) = (f - f * \rho_t)(x) + f * \rho_t(x), \tag{5.6}$$

$$\begin{aligned}(f - f * \rho_t)(x) &= \int_{\mathbb{R}^N} [f(x) - f(x-h)]\rho_t(h)dh = -\int_{\mathbb{R}^N} \Delta^1_{-h} f(x)\rho_t(h)dh \\ &= -\int_{\mathbb{R}^N} \Delta_{-th} f(x)\rho(h)dh,\end{aligned} \tag{5.7}$$

$$\begin{aligned}D(f * \rho_t)(x) =& \frac{1}{t}\int_{\mathbb{R}^N} f(x-h)(D\rho)_t(h)dh \\ =& \frac{1}{t}\int_{\mathbb{R}^N} [f(x-h) - f(x)](D\rho)_t(h)dh = \frac{1}{t}\int_{\mathbb{R}^N} \Delta_{-th} D\rho(h)dh,\end{aligned} \tag{5.8}$$

where we have used the fact that $\displaystyle\int_{\mathbb{R}^N} D\rho = 0$.

Assume that $\operatorname{supp}\rho \subset B(0,1)$. Combining (5.6)–(5.8), we find that, when $m = 1$, we have

$$\dot{K}(t,f) \le \|f - f * \rho_t\|_{L^p} + t\|D(f * \rho_t)\|_{L^p} \le \omega_p^1(t,f) + \|D\rho\|_{L^1}\omega_p^1(t,f), \tag{5.9}$$

and therefore the first inequality in (5.4) holds when $m = 1$.

When $m \ge 2$, some care is needed in the choice of the mollifier ρ, but one can still proceed as in (5.6)–(5.8). This is the demonstrated by the following lemma.

Lemma 5.7. Let φ be a mollifier. Set

$$a_j := \frac{(-1)^{j+1}}{m!}\binom{m}{j}, \ \forall\, 1 \le j \le m, \tag{5.10}$$

$$b_\ell := (-1)^{m-\ell}\binom{m}{\ell}, \ \forall\, 1 \le \ell \le m, \tag{5.11}$$

$$\rho := \sum_{1\le j,\ell\le m} a_j b_\ell \ell^m \varphi_{j\ell}. \tag{5.12}$$

Then, for every $f \in L^1_{loc}(R^n)$ and $\alpha \in \mathbb{N}^n$ with $|\alpha| = m$, we have

$$f(x) - f * \rho_t(x) = \frac{(-1)^m}{m!}\sum_\ell b_\ell \ell^m \int \Delta^m_{-th} f(x)\varphi_\ell(h)\, dh, \tag{5.13}$$

$$\partial^\alpha(f * \rho_t)(x) = t^{-m}\sum_j a_j \int \Delta^m_{-th} f(x)\partial^\alpha(\varphi_j)(h)\, dh. \tag{5.14}$$

The proof of Theorem 5.6 in [1] relies on a variant of Lemma 5.7, whose statement and proof seem less natural than the ones of our lemma.

Before proceeding to the proof of the lemma, let us note that it implies the first inequality in (5.4). Indeed, if $\operatorname{supp}\varphi \subset B(0,1/m^2)$, then $\operatorname{supp}\rho \subset B(0,1)$. For such φ, (5.13) and (5.14) imply that

$$\begin{aligned}\dot{K}(t^m,f) \leq& \|f - f*\rho_t\|_{L^p} + t^m\|D^m(f*\rho_t)\|_{L^p}\\ \leq& \left(\frac{1}{m!}\sum_{1\leq\ell\leq m}|b_\ell|\ell^m + \sum_{1\leq j\leq m}|a_j|\|D^m(\varphi_j)\|_{L^1}\right)\omega_p^m(t,f),\end{aligned}$$

whence the first inequality in (5.4).

We now proceed to the proof of Lemma 5.7. It will be more pedagogical not the start from the exact values of a_j, b_ℓ in (5.10)–(5.11), but from formula (5.12), with the coefficients a_j, b_ℓ to be determined in order that the desired identities (5.13)–(5.14) hold true.

Proof of Lemma 5.7. Clearly, the lemma holds for $m = 1$, so that we may assume that $m \geq 2$. Let a_j, $1 \leq j \leq m$, b_ℓ, $0 \leq \ell \leq m$, be constants to be chosen later, in order that ρ defined in (5.12) satisfies the (5.13)–(5.14).

Choice of b_ℓ, $1 \leq \ell \leq m$, such that (5.14) holds. Let $\alpha \in \mathbb{N}^N$ be such that that $|\alpha| = m$. Using the identity

$$\partial^\alpha((\varphi_{s\sigma})_t) = \sigma^{-m}t^{-m}(\partial^\alpha(\varphi_s))_{\sigma t},\ \forall\, s,\sigma,t>0,$$

and the fact that $\int \partial^\alpha\varphi = 0$, we find that (for every b_0)

$$\begin{aligned}\partial^\alpha(f*\rho_t)(x) =& t^{-m}\sum_j\sum_{\ell\geq 1} a_jb_\ell f*(\partial^\alpha(\varphi_j))_{\ell t}(x)\\ =& t^{-m}\sum_j a_j\int\sum_{\ell\geq 1} b_\ell f(x-\ell th)\partial^\alpha(\varphi_j)(h)\,dh\\ =& t^{-m}\sum_j a_j\int\sum_{\ell} b_\ell f(x-\ell th)\partial^\alpha(\varphi_j)(h)\,dh.\end{aligned}\tag{5.15}$$

We now let

$$b_\ell := (-1)^{m-\ell}\binom{m}{\ell},\ \forall\, 0\leq\ell\leq m.\tag{5.16}$$

With b_ℓ as in (5.16), (5.15) becomes

$$\partial^\alpha(f*\rho_t)(x) = t^{-m}\sum_j a_j\int\Delta^m_{-th}f(x)\partial^\alpha(\varphi_j)(h)\,dh\tag{5.17}$$

(by the binomial formula (5.1)), and thus for this choice of b_ℓ (5.14) does hold.

Choice of a_j, $1 \leq j \leq m$, such that (5.13) holds. For any a_0, we have

$$f*\rho_t(x) + \sum_\ell b_\ell\ell^m a_0 f(x) = \sum_\ell b_\ell\ell^m\int\sum_{j\geq 0}a_jf(x-jth)\varphi_\ell(h)\,dh.\tag{5.18}$$

Letting

$$a_j := C(-1)^{m-j}\binom{m}{j},\ \forall\, 0 \le j \le m, \tag{5.19}$$

(with C a constant to be fixed later), we obtain that

$$f * \rho_t(x) + (-1)^m C \sum_\ell b_\ell \ell^m f(x) = C \sum_\ell b_\ell \ell^m \int \Delta^m_{-th} f(x)\varphi_\ell(h)\, dh. \tag{5.20}$$

Assume now that $\sum_\ell b_\ell \ell^m \neq 0$. Then we choose

$$C := (-1)^{m+1} \frac{1}{\sum_\ell b_\ell \ell^m}. \tag{5.21}$$

For this choice of C (and the corresponding choice of a_j), we have

$$f(x) - f * \rho_t(x) - = -C \sum_\ell b_\ell \ell^m \int \Delta^m_{-th} f(x)\varphi_\ell(h)\, dh. \tag{5.22}$$

In order to complete the argument, we will prove that, indeed, $\sum_\ell b_\ell \ell^m \neq 0$ (and even find the value of this sum, and thus C).

Calculation of C. There exists constants $\alpha_0, \dots, \alpha_m$, with $\alpha_m = 1$, such that

$$X^m = \alpha_0 + \alpha_1 X + \sum_{2 \le j \le m} \alpha_j X(X-1)\cdots(X-j+1). \tag{5.23}$$

We have

$$\sum_\ell (-1)^\ell \binom{m}{\ell} = 0 \tag{5.24}$$

and, for $1 \le j \le m-1$,

$$\begin{aligned}\sum_\ell (-1)^\ell \binom{m}{\ell} \ell \cdots (\ell - j + 1) &= \sum_{j \le \ell \le m} (-1)^\ell \binom{m}{\ell} \ell(\ell-1)\cdots(\ell-j+1) \\ &= \sum_{j \le \ell \le m} (-1)^\ell (m-j+1)\cdots m \binom{m-j}{\ell-j} \\ &= (-1)^j (m-j+1)\cdots m \sum_{k \le m-j} (-1)^k \binom{m-j}{k} = 0.\end{aligned} \tag{5.25}$$

Combining (5.23)–(5.25), we find that

$$\begin{aligned}\sum_\ell (-1)^\ell \binom{m}{\ell} \ell^m &= \sum_\ell (-1)^\ell \binom{m}{\ell} \ell(\ell-1)\cdots(\ell-m+1) \\ &= (-1)^m \binom{m}{m} m(m-1)\cdots(m-m+1) = (-1)^m m!.\end{aligned} \tag{5.26}$$

From (5.26), we obtain that $\sum_\ell b_\ell \ell^m = m!$, and therefore $C = (-1)^{m+1}/m!$. From this, (5.17), and (5.22), we find that (5.14) holds. □

5.3 Extrapolation in fractional Sobolev Spaces

In this section, we denote

$$K_{N,p} := \left\{ \frac{1}{p} \int_{\mathbb{S}^{N-1}} |e_1 \cdot \nu|^p d\mathcal{H}^{N-1}(\nu) \right\}^{1/p},$$

where $e_1 := (1, 0, \dots, 0)$. We aim to present a proof of the following.

Theorem 5.8. Let $f \in W^{1,p}(\mathbb{R}^N)$, $1 \leq p < \infty$. We have that

$$\lim_{s \to 1^-} (1-s)^{1/p} |f|_{W^{s,p}} = K_{N,p} \|\nabla f\|_{L^p}. \tag{5.27}$$

On the other hand, for $f \in L^p(\mathbb{R}^N)$, we have the following:

$$\liminf_{s \to 1^-} (1-s)^{1/p} |f|_{W^{s,p}} < \infty \implies f \in W^{1,p}(\mathbb{R}^N), \text{ when } 1 < p < \infty\,; \tag{5.28}$$

$$\liminf_{s \to 1^-} (1-s) |f|_{W^{s,1}} < \infty \implies f \in BV(\mathbb{R}^N). \tag{5.29}$$

We will base our proof on the following lemma.

Lemma 5.9. Let $f \in W^{1,p}(\mathbb{R}^N)$. Then

$$T_p(t,f) := \sup_{|h| \leq t} \left\{ \frac{\|f(\cdot + h) - f(\cdot) - \nabla f(\cdot) \cdot h\|_{L^p}}{\|h\|} \right\} \xrightarrow[t \to 0^+]{} 0. \tag{5.30}$$

Before proceeding to the proof of Lemma 5.9 and Theorem 5.8, let us consider a proposition that is the analogue of Theorem 5.8, for the Sobolev norm (see Theorem 5.5) $\left\{ \displaystyle\int_0^\infty \frac{\omega_p^1(t,f)}{t^{sp+1}}\, dt \right\}$.

Proposition 5.10. Let $f \in L^p(\mathbb{R}^N)$. We have

$$\liminf_{t \to 0} \frac{\omega_p(t,f)}{t} < \infty \implies f \in W^{1,p}(\mathbb{R}^N), \text{ in the case where } 1 < p < \infty\,; \tag{5.31}$$

$$\liminf_{t \to 0} \frac{\omega_1(t,f)}{t} < \infty \implies f \in BV(\mathbb{R}^N). \tag{5.32}$$

Moreover, if $f \in W^{1,p}$, $1 \leq p < \infty$,

$$\limsup_{t \to 0^+} \frac{\omega_p(t,f)}{t} \leq \|\nabla f\|_{L^p}, \tag{5.33}$$

$$\|\nabla f \cdot \nu\|_{L^p} \leq \liminf_{t \to 0^+} \frac{\omega_p(t,f)}{t}, \text{ for any unit vector } \nu. \tag{5.34}$$

One proves (5.31) and (5.32) by: 1) showing these results hold for functions in C^1 ; 2) by a compactness argument applied on approximating sequences of the form $f * \rho_n$. We omit the proof. On the other hand, one easily deduces the estimates (5.33) and (5.34) from Lemma 5.9. We do not enter into the details of these proofs, and move on to the proof of Lemma 5.9.

Proof of Lemma 5.9. We only need to show that (5.30) holds for $f \in C_c^2(\mathbb{R}^N)$.
This comes from a general observation that we often use in Sobolev estimates of this type.

Observation 5.11. Let $(X,d),(Y,\delta)$ be metric spaces and $F:\mathbb{R}^+\times X\to Y$ and $g:X\to Y$ be two functions. If F and g are such that :

1. g is continuous and the family $(F(t,\cdot))_{t>0}$ is equicontinuous.
2. $\lim_{t\to 0^+}F(t,x)=g(x)$ for all $x\in D$, a dense subset of X.

Then, one gets that $\lim_{t\to 0^+}F(t,x)=g(x)$ for all $x\in X$.

We would like to apply this observation to $F=T_p$ and $g=0$ that are defined on $X=W^{1,p}(\mathbb{R}^N)$ and with values in $Y=\mathbb{R}$, with the set $D=C_c^2(\mathbb{R}^N)$, which is dense in $W^{1,p}(\mathbb{R}^N)$. We are left with verifying that, in this case, 1. and 2. in Observation 5.11 hold.

1. We indeed have that $(T_p(t,\cdot))_t$ is equicontinuous. One finds that, for all $f,g\in W^{1,p}(\mathbb{R}^N)$,

$$|T_p(t,f)-T_p(t,g)|\leq 2\|\nabla(f-g)\|_{L^p}$$

since, for any $h\neq 0$, we have

$$\frac{\|f(\cdot+h)-f(\cdot)-\nabla f(\cdot)\cdot h\|_{L^p}}{\|h\|}\leq\frac{\|g(\cdot+h)-g(\cdot)-\nabla g(\cdot)\cdot h\|_{L^p}}{\|h\|}+2\|\nabla(f-g)\|_{L^p}$$

and the same for g and f exchanged, because of the well-known estimate

$$\|\zeta(\cdot+h)-\zeta(\cdot)\|_{L^p}\leq\|\nabla\zeta\|_{L^p}\|h\|\text{ for any }\zeta\in W^{1,p}.$$

2. On the other hand, we have that $T_p(t,f)\xrightarrow[t\to 0^+]{}0$ for all $f\in C_c^2(\mathbb{R}^N)$. This fact is clear since, if $f\in C_c^2(\mathbb{R}^N)$

$$\begin{aligned}\frac{\|f(\cdot+h)-f(\cdot)-\nabla f(\cdot)\cdot h\|_{L^p}}{\|h\|}&=\frac{\|\int_0^1(1-t)D^2f(\cdot+th)(h,h)dt\|_{L^p}}{\|h\|}\\&\leq\left\{\int_0^1(1-t)\|D^2f(\cdot)\|_{L^p}dt\right\}\|h\|\end{aligned}$$

by applications of Minkowski's and Cauchy-Schwarz inequalities. Thus, one finally finds, by passing to the supremum in the previous inequality that

$$\begin{aligned}&T_p(t,f)\leq C_ft,\text{ for all }t>0\\&T_p(t,f)\xrightarrow[t\to 0^+]{}0\end{aligned}$$

for all $f\in C_c^2(\mathbb{R}^N)$. Therefore, one can apply Observation 5.11 to this particular case, and conclude that Theorem 5.9 holds. □

The relevance of Lemma 5.9 with respect to the proof of (5.27) is based on the following observation.

Observation 5.12. For any $0<r<1$,

$$0\leq\int_{|h|\geq r}\frac{\|f(\cdot+h)-f(\cdot)\|_{L^p}^p}{\|h\|^{sp+N}}dh\leq 2^p\beta_n\|f\|_{L^p}\int_r^{+\infty}\frac{1}{s^{sp+1}}ds=\frac{2^p\beta_n\|f\|_{L^p}}{(sp)r^{sp}}.$$

Thus, one finds that $(1-s)\displaystyle\int_{|h|\geq r}\frac{\|f(\cdot+h)-f(\cdot)\|_{L^p}^p}{\|h\|^{sp+N}}dh\to 0$ when $s\to 1^-$.

Observation 5.12, combined with Lemma 5.9, shows that we only need to determine the following limit (see details in the proof of Theorem 5.8).

Lemma 5.13. For any $\delta > 0$, and for any $f \in \dot{W}^{1,p}(\mathbb{R}^N)$, one has that

$$\lim_{s\to 1^-} (1-s)^{1/p} \left\{ \int_{|h|<\delta} \frac{\|\nabla f \cdot h\|_{L^p}^p}{\|h\|^{sp+N}} dh \right\}^{1/p} = K_{N,p} \|\nabla f\|_{L^p}$$

Proof of Lemma 5.13. We have

$$\int_{|h|<\delta} \frac{1}{\|h\|^{sp+N}} \int_{\mathbb{R}^n} |\nabla f(x)\cdot h|^p dx dh = \int_{\mathbb{R}^n} \int_0^\delta \frac{1}{r^{sp+1}} \int_{\mathbb{S}^{N-1}} |\nabla f(x)\cdot r\nu|^p d\mathcal{H}^{N-1}(\nu) dr dx$$

$$= \int_{\mathbb{R}^N} |\nabla f(x)|^p \int_0^\delta \frac{1}{r^{(s-1)p+1}} \int_{\mathbb{S}^{N-1}} |e_1\cdot\nu|^p d\mathcal{H}^{N-1}(\nu) dr dx,$$

where we factorized r and the norm of $\nabla f(x)$ out of the scalar product, and then used the fact that for every unit vectors V and e,

$$\int_{\mathbb{S}^{N-1}} |V\cdot\nu|^p d\mathcal{H}^{N-1}(\nu) = \int_{\mathbb{S}^{N-1}} |e\cdot\nu|^p d\mathcal{H}^{N-1}(\nu),$$

because of the rotational invariance of the Hausdorff measure on the sphere. One finally finds that

$$(1-s)^{1/p} \left\{ \int_{|h|<\delta} \frac{\|\nabla f(\cdot)\cdot h\|_{L^p}^p}{\|h\|^{sp+N}} dh \right\}^{1/p} = \frac{(1-s)^{1/p}\delta^{1-s}}{((1-s)p)^{1/p}} \|\nabla f\|_{L^p} \Big(\int_{S^{N-1}} |e_1\cdot\nu|^p d\mathcal{H}^{N-1}(\nu) \Big)^{1/p}$$

$$\to K_{N,p} \|\nabla f\|_{L^p}$$

as $s \to 1^-$, with $K_{N,p} = \left\{ \frac{1}{p} \int_{S^{N-1}} |e_1\cdot\nu|^p d\mathcal{H}^{N-1}(\nu) \right\}^{1/p}$. □

Proof of Theorem 5.8. **Proof of (5.28) and (5.29), using (5.31) and (5.32):**

Consider $f \in L^p(\mathbb{R}^N)$ such that $\liminf_{s\to 1^-} (1-s)^{1/p} |f|_{W^{s,p}} < \infty$. We have that (see Theorem 5.5)

$$(1-s) \int_0^\infty \omega_p^1(t,f)^p \frac{dt}{t^{sp+1}} \le \frac{c(1-s)}{s^p} |f|_{W^{s,p}}^p \, dh, \text{ for all } 0<s<1. \tag{5.35}$$

On the other hand, one can apply Theorem 4.12 to the function $\omega_p(\cdot, f)$. Passing to the $\liminf_{s\to 1^-}$ in (5.35), one finds that

$$\liminf_{t\to 0^+} \left\{ \frac{\omega_p^1(t,f)}{t} \right\}^p \le \liminf_{s\to 1^-} (1-s) \int_0^\infty \omega_p^1(t,f)^p \frac{dt}{t^{sp+1}}$$

$$\le \liminf_{s\to 1^-} \frac{c(1-s)}{s^p} |f|_{W^{s,p}}^p < \infty.$$

Therefore, we have that $\liminf_{t\to 0^+} \frac{\omega_p^1(t,f)}{t} < \infty$. We can conclude, by (5.31) and (5.32), that f is in $W^{1,p}$ if $p>1$, and in BV if $p=1$.

Proof of (2.7): In order to show that $\limsup_{s\to 1^-} (1-s)^{1/p} |f| \le K_{N,p} \|\nabla f\|_{L^p}$, we will use the following inequality.

We have, for all $h \in \mathbb{R}^N$,

$$\begin{aligned}\|f(\cdot+h)-f(\cdot)\|_{L^p} &\le \|\nabla f\cdot h\| + \|f(\cdot+h)-f(\cdot)-\nabla f(\cdot)\cdot h\|_{L^p}\\ &\le \|\nabla f\cdot h\|_{L^p} + \|h\|\, T_p(\|h\|, f).\end{aligned} \tag{5.36}$$

Let $\epsilon > 0$. There exists $0 < t_0 < 1$ such that $0 \le T_p(z,f) \le \epsilon$ for all $0 < z \le t_0$ (Lemma 5.9). We have

$$\begin{aligned}(1-s)^{1/p}|f|_{W^{s,p}} \le (1-s)^{1/p}\Bigg\{ &\left[\int_{\|h\|\le t_0} \frac{\|f(\cdot+h)-f(\cdot)\|_{L^p}^p}{\|h\|^{sp+N}}dh\right]^{\frac{1}{p}}\\ &+\left[\int_{\|h\|>t_0} \frac{\|f(\cdot+h)-f(\cdot)\|_{L^p}^p}{\|h\|^{sp+N}}dh\right]^{1/p}\Bigg\}\end{aligned}$$

and we find, using (5.36), that

$$\begin{aligned}(1-s)^{1/p}|f|_{W^{s,p}} \le (1-s)^{1/p}\Bigg\{ &\left[\int_{\|h\|\le t_0} \frac{\|\nabla f(\cdot)\cdot h\|^p}{\|h\|^{sp+N}}dh\right]^{1/p}\\ &+\left[\int_{\|h\|\le t_0} \frac{T_p(f,\|h\|)}{\|h\|^{N-(1-s)p}}dh\right]^{1/p}\\ &+\left[\int_{\|h\|>t_0} \frac{\|f(\cdot+h)-f(\cdot)\|_{L^p}^p}{\|h\|^{sp+N}}dh\right]^{1/p}\Bigg\}.\end{aligned}$$

Since $0 \le T_p(t,f) \le \epsilon$ for all $0 < z \le t_0$, we get that

$$\begin{aligned}(1-s)^{1/p}|f|_{W^{s,p}} \le (1-s)^{1/p}\Bigg\{ &\left[\int_{\|h\|\le t_0} \frac{\|\nabla f(\cdot)\cdot h\|^p}{\|h\|^{sp+N}}dh\right]^{1/p} + \epsilon\left(\frac{\beta_N t_0}{(1-s)p}\right)^{1/p}\Bigg\}\\ &+(1-s)^{1/p}\left[\int_{\|h\|>t_0} \frac{\|f(\cdot+h)-f(\cdot)\|_{L^p}^p}{\|h\|^{sp+N}}dh\right]^{1/p}\end{aligned} \tag{5.37}$$

Finally, we may pass to the $\limsup_{s\to 1^-}$ in (5.37), and apply Observation 5.12 and Lemma 5.13, to find that, for all $\epsilon > 0$,

$$\begin{aligned}\limsup_{s\to 1^-}(1-s)^{1/p}|f|_{W^{s,p}} &\le K_{N,p}\|\nabla f\| + \epsilon\left(\frac{\beta_N t_0}{p}\right)^{1/p}\\ &\le K_{N,p}\|\nabla f\| + \epsilon\left(\frac{\beta_N}{p}\right)^{1/p} \text{ (since } t_0 < 1).\end{aligned}$$

Considering the limit as $\epsilon \to 0$ in the last estimate, we get to the desired conclusion.

The proof of $\liminf_{s\to 1^-}(1-s)^{1/p}|f|_{W^{s,p}} \ge K_{N,p}\|\nabla f\|_{L^p}$ is similar. One uses, instead of (5.36), the following inequality.

$$\begin{aligned}\|f(\cdot+h)-f(\cdot)\|_{L^p} &\ge \|\nabla f\cdot h\| - \|f(\cdot+h)-f(\cdot)-\nabla f(\cdot)\cdot h\|_{L^p}\\ &\ge \|\nabla f\cdot h\|_{L^p} - \|h\|\, T_p(\|h\|, f).\end{aligned} \tag{5.38}$$

□

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